

NEURAL BELLMAN OPERATORS

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We study reusable policies for changes in economic contracts and transition laws. Neural Bellman Operators separate proposal, evaluation, improvement, and certification. For affine finite-horizon transition mixtures, a signed endpoint correction compresses a sharper count-information upper value; feasible-policy Bernstein representations deliver regional welfare and decision certificates with an explicit policy-bank error account. In a portfolio economy with adjustable risk aversion, the certified decision concerns a positive-duration initial position. We embed the operating economy in a participation contract and characterize its signing grant, service-procurement condition, and statewide enforcement frontier. Adjustment weakly reduces the compensation and enforcement needed to operate, but its relative value across position classes can reverse under voluntary surrender. A finite-charge region implements the original opposite-position comparison. An exact first-date risky-share reduction and permission shadow values explain how long and short limits move both indifference boundaries. Common-cardinal-tilt robustness, dynamic feature counterfactuals, and wealth-settlement perturbations retain distinct economic roles. Streaming chord and count are compared on the same decision, including common construction and validation costs; mesh-only and exact structural countercomparisons remain visible. Recursive utility, temporal selves, and persistent capacity competition extend the framework. Computational precision is linked to the price and scope of an economic contract, rather than equated with a training residual.

KEYWORDS: Dynamic programming, Neural policy iteration, Recursive utility, Endogenous preferences, Stochastic control, Computational economics.

1. INTRODUCTION

Economic counterfactuals require more than a numerical solution at a single calibration. A researcher changes an operating benefit, an adjustment cost, or the distribution of a financial shock and asks whether a policy response reflects an economic mechanism rather than approximation error. Repeating an optimization at every parameter value is one response. Reusing an approximate policy is another, but it is useful only when its welfare loss and the scope of its validity can be evaluated. This paper studies that problem for dynamic economies in which control changes both financial resources and the agent's preferences.

An operating mandate also requires an economic account of its obligations. We introduce voluntary participation, an externally funded signing grant, finite enforcement capacity, and a right to surrender after a minimum term. Preference adjustment can lower both the compensation required to operate and the enforcement capacity needed to implement continued operation. Its relative value across initial position classes can nevertheless reverse when the surrender margin changes. We derive that distinction and map the two investment-permission boundaries using an exact first-date breakpoint reduction. These results tie a certified risk comparison to the price of the contract that makes it relevant.

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Neural Bellman Operators (NBO) separate policy proposal, evaluation, improvement, and certification. A critic represents continuation value, an actor proposes a feasible action, an improvement calculation checks that action, and an independent evaluation measures the induced policy. We use this separation to develop a counterfactual method with an explicit distinction between changes in rewards and changes in transition laws. The certificate, rather than a training objective or a preferred approximation architecture, determines which economic conclusions are warranted.

The first distinction is mathematical. For fixed transitions, the value of a policy is affine in reward coefficients. Policy caches and optimistic linear support exploit precisely this structure. Once a parameter changes the transition law at every date, even a fixed policy can have a nonconvex value in that parameter. We identify the cross-operator term that makes an uncorrected endpoint chord fail and construct a positive backward correction yielding an upper value throughout a parameter interval. Its additional error is second order in interval width under explicit Lipschitz conditions. A matched comparison shows that a localized count-information upper value is sharper, whereas the correction compresses triangular horizon dependence into a linear number of action-wide kernel applications. We give a total coefficient-error bound and a finite-refinement guarantee that remain valid when optimal actions switch. Independently evaluated policy values have a finite Bernstein representation for an affine mixture of transition laws. Combining their maximum with the corrected upper value and statewise policy switching gives a welfare certificate at every state and date over a continuum of transition counterfactuals. The result also admits approximate neural endpoint values after a certified residual correction.

The second distinction is economic. In a portfolio economy with adjustable risk aversion and first-exit liquidation, a constant operating benefit m and a liquidation increment ℓ affect behavior through the contrast $d = m - \rho\ell$. The annuity identity explains which contract changes matter, but not whether a portfolio reversal is caused by preference formation. We partition policies by the sign of the risky position held over a positive-length initial rebalancing interval. The slope of a regular indifference locus with respect to adjustment cost equals the difference in adjustment effort between the two risk classes divided by their difference in discounted duration. A separate option-value decomposition compares deliberate preference adjustment with the otherwise identical economy in which adjustment is unavailable. It identifies the quantity that must change the ranking, but an identity alone does not predict that change. We derive a primitive sufficient condition from consumption exposure and the curvature of CRRA utility with respect to its preference index. It predicts a strictly positive risk-specific adjustment premium over an entire probability-cost family. In the full finite stopped economy, class-specific tensor certificates then establish opposite risk rankings with and without adjustment over a joint region of contract and transition parameters.

The finite implementation retains consumption, risky investment, costly preference adjustment, stochastic preference and wealth shocks, their covariance, and the original settlement rule. It includes the union of both historical action meshes and the frozen neural proposals. The reward-transfer bank retains its all-state, all-date 10^{-3} welfare certificate and its separate first-action sign exclusions. The new uncertainty experiment changes the probability of two one-step shock laws, each with its own killed transition kernel, and certifies their entire mixture interval. This changes the dynamics; it is not a relabeling of a reward coefficient. The finite-chain mixture is distinguished from changing a Brownian correlation parameter without a discretization bound.

The economic interpretation is developed further by specifying the decision calendar and settlement lottery, varying its wealth granularity with both adjustment regimes

recomputed, and separating the resulting movement of a risk boundary from the relative adjustment option. We retain the original region, including the loss of its no-adjustment sign at a finer-settlement corner. A common nested region is additionally certified for each of three finite protocols. A preference over preference states supplies the cardinal primitive not identified by a CRRA index alone. We prove a uniform option bound under bounded common cardinal tilts and then use full dynamic reoptimization to distinguish that primitive from operating-duration, liquidation, and covariance incentives. These results concern explicit economic experiments rather than a sign assigned to an unspecified continuous-time limit.

The computational comparisons assign credit to the object doing the work. We execute an exact-oracle scalar optimistic-linear-support comparison, distinguish an initial-distribution objective from simultaneous statewise coverage, and remove all neural actions from the lower policy bank while retaining the original full-menu upper benchmark. In the resource economy, we add an analytically assembled, probability-preconditioned parametric quadratic program that reuses the same structure across fresh queries. It is substantially faster than the learned arm in this short quadratic model. This comparison changes the inference from the earlier crossover: nonlinear continuation can outperform the fitted quadratic approximation without outperforming the exact reusable quadratic program. The transition-law experiments compare upper oracles on identical anchors, feasible policies, and subdivision cells, charging shared setup and evaluation work to each arm. They also measure autonomous dyadic refinement from empty policy caches. The compressed and count constructions occupy different points on a precision-work frontier; the rectangular relaxation is a further regional comparator. The economic use is a decision-specific region certificate rather than a generic welfare bound placed beside a separately optimized risk bracket. A new streaming comparison charges both upper methods for the same model construction, full-menu endpoint optimization, feasible-policy evaluation, and independent replay at that decision target. It also tests a mesh-only feasible lower bank against the unchanged full target. This comparison attributes a gain to compression without treating learned proposals as indispensable.

The general error analysis and the broader economic extensions remain part of the argument. A finite-horizon theorem distinguishes evaluation, improvement, boundary, and transition errors. Exact policy improvement provides a conditional benchmark; counterexamples exclude identification of attracting actor stationary points with globally optimal policies. Merton and Epstein–Zin examples test actual evaluation and control updates. Sophisticated temporal selves require a different improvement criterion from ordinary continuation evaluation. Persistent capacity competition tests unilateral dynamic deviations at every date and state. These exercises retain their separate economic purposes rather than becoming an aggregate count of successes.

1.1. *Related Literature and the Contribution*

Policy improvement and approximate dynamic programming precede neural approximation. [Jacka and Mijatović \(2015\)](#) study continuous-time improvement under explicit control hypotheses. The nearest neural approaches also combine continuation approximation and policy improvement: fixed-policy PDE evaluation in [Kim, Kim, Kim, and Cho \(2025, 2026\)](#), operator learning in [Lee and Kim \(2024\)](#), martingale evaluation in [Cai, Fang, and Zhou \(2025\)](#), and direct control approximation in [Han and E \(2016\)](#). Neural evaluation, separate parameter blocks, and avoidance of a closed-form maximizer are

not claims of priority here. Our operator assumptions and error accounts specify when such computations support the economic use of a policy.

Successor features and generalized policy improvement (Barreto et al., 2017), policy switching (Chang, 2021), and policy-cache upper bounds (Nemecek and Parr, 2021) provide the reward-transfer foundations. Alegre, Bazzan, and da Silva (2022) combine optimistic linear support with successor features and generalized policy improvement. Their work is a necessary comparator: selecting an unsatisfactory reward weight, solving there, and adding its policy is not a new algorithm merely because the reward weight is a contract. We prove below that, with exact statewise endpoint optima and one policy per scalar anchor, our adjacent-policy envelope equals the full scalar-OLS policy envelope. We measure the cost of strengthening an initial-distribution guarantee to simultaneous statewise coverage, but do not present that strengthening alone as a new transfer principle. Our switching rule also differs from full action-value generalized policy improvement; the comparison does not identify the two operators.

The additional transfer result concerns *changing transition laws*, outside the fixed-feature reward correspondence. Lipschitz dependence of value on rewards and transitions is established in the changing-environment literature (Csáji and Monostori, 2008). Theorem 10 instead gives an explicit endpoint-supersolution correction involving the product of a kernel difference and a continuation-value difference. Its implementation is paired with exact feasible-policy polynomials, a statewise welfare bound, and an independently computed information-relaxation upper value. The latter is an instance of the general approach of Brown, Smith, and Sun (2010), not a new duality theorem. Regional verification and parameter lifting (Quatmann, Dehnert, Jansen, Junges, and Katoen, 2016, Heck, Quatmann, Spel, Katoen, and Junges, 2025) are also direct comparators for repeated transition parameters; we distinguish their dependence relaxations and implement a finite-horizon rectangular upper recursion on the same economic target. Section 10 gives the date-augmented reward-MDP correspondence, the order of action and endpoint information, the treatment of killed mass and terminal rewards, and the precise role of signed compression. The economic switching-locus result uses the envelope logic of Milgrom and Segal (2002); the envelope identity organizes the calculation but does not itself establish a duration ordering. The additional economic result is a primitive consumption-exposure condition and an integrated class-value certificate, not a claim to a new envelope theorem.

The broader computational literature includes recursive and projection methods (Stokey, Lucas, and Prescott, 1989, Judd, 1998), sparse grids (Brumm and Scheidegger, 2017), and neural differential-equation solvers (Raissi, Perdikaris, and Karniadakis, 2019, Han, Jentzen, and E, 2018). Our input-convex critic follows Amos, Xu, and Kolter (2017). Riccati recursion and a reusable convex quadratic program are the appropriate structural comparators for the corresponding models. We retain both, including results unfavorable to learned deployment. Thus the paper’s incremental case rests on the controlled compression of transition-law upper information, the resulting precision-work and anchor-demand comparisons, and a certified preference-adjustment effect on economic choice. The architecture proposes policies; theorems and independent class evaluation determine which counterfactual conclusions those policies support.

2. ECONOMIC CONTROL AND SEPARATED BELLMAN OPERATORS

2.1. *The Controlled Problem*

Let $S_t = (u_t, X_t)$ be a Markov state and let $a_t = (\theta_t, b_t)$ collect internal and external controls. On a specified domain D , the controlled state satisfies

$$dS_t = \mu(t, S_t, a_t) dt + B(t, S_t, a_t) dW_t, \quad a_t \in A(t, S_t), \quad (1)$$

where W is a vector of independent Brownian motions and $A(t, s)$ is the feasible correspondence. Correlation is represented in B . More generally, a continuous martingale with covariance density $Q(t, s, a)$ gives covariance $\sigma Q \sigma^\top$. A predictable covariance process not determined by the declared state does not, merely by notation, define the same Markov problem; the state must be augmented or the formulation changed.

For additive preferences and stopping time $\tau \leq T$, policy evaluation is

$$J^\pi(t, s) = \mathbb{E}_{t,s}^\pi \left[\int_t^\tau e^{-\rho(v-t)} r(v, S_v, \pi_v) dv + e^{-\rho(\tau-t)} g(\tau, S_\tau) \right]. \quad (2)$$

The stopping rule and payoff are economic primitives. In an exit problem, τ is the first exit or terminal date. Reflection instead requires a regulator and a boundary operator involving its direction; state constraints require viability. Neither interpretation is supplied by an unreported numerical projection.

For recursive preferences, evaluation is first defined for each admissible policy by

$$-dY_t^\pi = f(t, S_t, \pi_t, Y_t^\pi) dt - Z_t^\pi dW_t, \quad Y_\tau^\pi = g(\tau, S_\tau). \quad (3)$$

The driver may incorporate a preference-adjustment cost. Existence, integrability, comparison, and any domain restriction on its value argument are imposed before taking a supremum over policies. Lipschitz drivers on a finite horizon provide a standard admissible class; the Epstein–Zin specification below uses its own positive-domain and transversality conditions. Calling Y an argument of utility does not itself establish well-posedness.

When a classical Markov representation is available, write

$$\mathcal{H}^a[v] = f(t, s, a, v) + \mu(t, s, a)^\top \nabla v + \frac{1}{2} \text{tr} \{ B(t, s, a) B(t, s, a)^\top \nabla^2 v \}. \quad (4)$$

For additive utility, $f = r - \rho v$. The evaluation equation is $v_t + \mathcal{H}^\pi[v] = 0$ with the specified boundary operator, while the control equation replaces $\mathcal{H}^\pi$ by $\sup_{a \in A} \mathcal{H}^a$. Boundary data, admissibility, and comparison are part of these statements, not consequences of a neural parameterization.

2.2. *Evaluation, Proposal, and Improvement*

Let v_ξ be a critic and a_ω a feasible actor. A differential evaluation step freezes the policy and minimizes

$$L_E(\xi; \pi) = \mathbb{E}_v \left[(\partial_t v_\xi + \mathcal{H}^\pi[v_\xi])^2 \right] + \lambda_B \mathbb{E}_{v_B} \left[|\mathcal{B}v_\xi - g_B|^2 \right]. \quad (5)$$

Here $\mathcal{B}$ is the actual boundary operator. A Dirichlet discrepancy is appropriate for the stopped problem used below, not for an unspecified reflected diffusion. Hard

architectures can impose the boundary exactly, although they do not eliminate interior approximation error.

The actor’s Hamiltonian proposal solves, approximately,

$$\max_{\omega} \mathbb{E}_{v_I} [\mathcal{H}^{\omega} \{ \text{sg}(v_{\xi}), \text{sg}(\nabla v_{\xi}), \text{sg}(\nabla^2 v_{\xi}) \}]. \quad (6)$$

The notation sg freezes the critic’s parameter block. Coefficients depending on the action, including diffusion loadings, retain their action derivatives. Evaluation does not update actor parameters, and actor improvement does not update critic parameters. For a game, rival controls are also frozen in a player’s improvement step. These restrictions concern the computational graph, not just the labels attached to two loss functions.

A bounded activation can enforce a box constraint. Shared resource constraints require a different map. For example, $a_i = B_0 \exp(z_i) / (\sum_{j=1}^d \exp(z_j) + \exp(z_0))$ gives $a_i \geq 0$ and $\sum_i a_i \leq B_0$, with z_0 representing slack. General convex sets can use projection; a penalty alone is an approximation whose violations must be measured. An arbitrary policy sampling measure does not turn (6) into the gradient of lifetime welfare. Such an interpretation requires occupancy or adjoint weighting and the relevant interchange conditions.

For an implemented transition rule, it is often preferable to use a finite-step operator. At dates $n = 0, \dots, N - 1$, define

$$(\mathcal{T}_n^a v)(s) = R_n(s, a) + \int v(s') K_n(ds' | s, a), \quad \mathcal{T}_n v = \sup_{a \in A_n(s)} \mathcal{T}_n^a v. \quad (7)$$

The nonnegative kernel includes discount and survival. A stopped payoff is included in R_n ; it is not continued at a clipped interior state. The experiments with nonlinear critics proceed backward: improve the actor against the frozen next-date critic, then fit the current critic to the evaluated target of the selected action. This is a finite-horizon separated Bellman update, not a simulation of a joint scalar loss.

In this semigroup implementation, freezing the continuation network’s parameters must not remove its derivative with respect to the next state. The actor influences s' through the transition, so the composition $v_{n+1}(F(s, a, \varepsilon))$ must retain that input derivative. Our tests compare this actor gradient with finite differences and separately check the differential Hamiltonian graph. The distinction prevents a stop-gradient convention appropriate for a Hamiltonian jet from being incorrectly imposed on a finite-step transition.

2.3. The Exact Policy-Improvement Benchmark

The following proposition states the exact object to which neural approximation is compared. Let $\mathcal{V}$ be a classical value space whose norm controls v , v_t , ∇v , $\nabla^2 v$, and the boundary traces required by $\mathcal{B}$. Including the time derivative is essential to pass the evaluation equation to a limit.

ASSUMPTION 1—Exact evaluation and improvement: *The policy class Π is compact in a stated uniform topology. Evaluation $E : \Pi \rightarrow \mathcal{V}$ is uniquely defined, continuous, and has precompact image. A selector I attains the pointwise Hamiltonian supremum, maps Π into itself, and is continuous at accumulation points of the policy sequence. Comparison and verification hold for the specified generator, recursive driver, boundary operator, and value class. Evaluated values are bounded above by the optimal value.*

PROPOSITION 1—Exact separated policy improvement: *Under Assumption 1, let $\pi_{j+1} = I(\pi_j)$ and $v_j = E(\pi_j)$. The evaluated values increase to a common limit in $\mathcal{V}$. Every accumulation policy p satisfies $E(p) = E(I(p)) = \bar{v}$, and $\bar{v}$ solves the maximized boundary-value problem. If comparison identifies the value, p is optimal.*

The important limit is equality of values, not equality of successive policies. Along $\pi_{j_k} \rightarrow p$, continuity gives $E(p) = \bar{v}$ and $E(I(p)) = \bar{v}$ from the shifted sequence. The evaluation equation for $I(p)$ at that same value is the maximized equation. The supplement proves this step in detail. Compactness and closure under an exact selector are substantial hypotheses; they are not automatically properties of a finite neural class. Proposition 1 is not an optimizer convergence theorem.

3. APPROXIMATION ERROR AND FEASIBLE IMPROVEMENT

3.1. An Error Account for Policy Value

Let $\mathcal{T}_n^a$ be the target evaluation operator and $\tilde{\mathcal{T}}_n^a$ the implemented operator. For additive preferences, these have the form (7). For recursive preferences, the following result also permits nonlinear evaluation maps, provided they are order preserving and Lipschitz in the continuation value. We impose those properties on the relevant value domain rather than presume them for every recursive aggregator.

ASSUMPTION 2—Finite-horizon operators: *For every feasible a , the target operator $\mathcal{T}_n^a$ is order preserving and satisfies $\|\mathcal{T}_n^a v - \mathcal{T}_n^a w\|_\infty \leq \gamma_n \|v - w\|_\infty$. The same bound holds for $\tilde{\mathcal{T}}_n = \sup_a \mathcal{T}_n^a$; the supremum is well defined. Terminal values and the selected policy values exist and are bounded. The numerical errors defined below are finite. The constants γ_n may exceed one for a finite-horizon recursive problem.*

For learned values $\hat{v}_n$ and a feasible policy π_n , define

$$\epsilon_n = \left\| \hat{v}_n - \tilde{\mathcal{T}}_n^{\pi_n} \hat{v}_{n+1} \right\|_\infty, \quad (8)$$

$$\delta_n = \left\| \tilde{\mathcal{T}}_n \hat{v}_{n+1} - \tilde{\mathcal{T}}_n^{\pi_n} \hat{v}_{n+1} \right\|_\infty, \quad (9)$$

$$\kappa_n = \sup_{s,a} \left| \tilde{\mathcal{T}}_n^a \hat{v}_{n+1}(s) - \mathcal{T}_n^a \hat{v}_{n+1}(s) \right|, \quad \eta = \|\hat{v}_N - g_N\|_\infty. \quad (10)$$

The first quantity is evaluation error; the second is a payoff improvement gap, not an action distance. The third compares operators on the actual learned continuation. It includes transition, quadrature, interpolation, and stopped-payoff errors when those are present. Let $\Gamma_{n,j} = \prod_{i=n}^{j-1} \gamma_i$, with an empty product equal to one.

THEOREM 2—Policy-value error account: *Under Assumption 2, the target policy value J_n^π satisfies*

$$\|\hat{v}_n - J_n^\pi\|_\infty \leq \sum_{j=n}^{N-1} \Gamma_{n,j} (\epsilon_j + \kappa_j) + \Gamma_{n,N} \eta. \quad (11)$$

For the target optimal value V_n^* ,

$$0 \leq V_n^*(s) - J_n^\pi(s) \leq \sum_{j=n}^{N-1} \Gamma_{n,j} (2\epsilon_j + \delta_j + 2\kappa_j) + 2\Gamma_{n,N} \eta. \quad (12)$$

The proof is backward error propagation, but the placement of the errors matters. The policy-evaluation recursion contributes $\epsilon_n + \kappa_n$. The optimized recursion contributes $\epsilon_n + \delta_n + \kappa_n$. Adding the resulting bounds yields (12); separately inflating an actor gap by operator error would produce an unnecessarily loose coefficient. For additive stopped control with exact discount, $\gamma_n \leq e^{-\rho \Delta_n}$. A boundary-payoff error at an intermediate exit belongs in the relevant one-step operator error, not only in the terminal error at N .

There is an analogous differential statement. If a classical candidate v has boundary discrepancy at most η , fixed-policy residual at most $\epsilon(t)$, and Hamiltonian gap at most $\delta(t)$, Itô's formula and verification give

$$V^*(t, s) - J^\pi(t, s) \leq 2\eta + \int_t^T e^{-\rho(q-t)} \{2\epsilon(q) + \delta(q)\} dq. \quad (13)$$

The stopped martingale and comparison conditions are stated in Supplement S.1. This is a quantitative link to a differential critic, not a claim that sampled residuals provide the required supremum bounds.

A finite test set by itself provides a lower observation of a supremum, not an upper certificate. With a known Lipschitz constant L and an h -cover, a residual's uniform magnitude is bounded by its largest tested magnitude plus Lh . Without such regularity and coverage information, we report held-out errors as diagnostics. The same distinction applies to critic derivatives, boundary layers, and off-policy states. Network approximation of a smooth function does not convert a sample average into a uniform error bound.

3.2. Two Improvement Safeguards

Let $Q_n(s, a) = \tilde{T}_n^a \hat{v}_{n+1}(s)$. For a finite feasible set $C_n(s)$ and neural proposal $a_\omega(s)$, the candidate safeguard selects

$$\pi_n(s) \in \arg \max_{a \in C_n(s) \cup \{a_\omega(s)\}} Q_n(s, a). \quad (14)$$

It weakly improves the proposal's evaluated target at every queried state. If $C_n(s)$ is an h_a -cover of the feasible set and Q_n has a verified action-Lipschitz constant L_a , its remaining improvement gap is at most $L_a h_a$. Without that information, a finite comparison is not a continuum optimality certificate. In the portfolio computation we use the safeguard and report the remaining independently evaluated feasible deviations.

For convex cost minimization, a stronger check is available. Let $F_s(a)$ be differentiable and convex on a compact convex feasible set $A(s)$. Define

$$g_s(a) = \nabla F_s(a)^\top a - \min_{b \in A(s)} \nabla F_s(a)^\top b. \quad (15)$$

Convexity implies $0 \leq F_s(a) - \min_b F_s(b) \leq g_s(a)$. For $A = \{a \geq 0 : \sum_i a_i \leq B_0\}$, the linear minimum is $B_0 \min\{0, \min_i \partial_i F_s(a)\}$. The computation therefore obtains an improvement certificate without a global search over neural parameters. The actor is a proposal; projected improvement is stopped according to (15).

In the resource application, the continuation critic is convex by construction, so the one-period criterion is strongly convex in the action. Its neural representation uses

nonnegative coefficients on squared-ReLU features together with a convex quadratic term and a free affine term. This architecture justifies the convex check for the *learned* continuation criterion. It does not assume that its continuation value is exact. An independent scenario-tree calculation separately assesses the resulting policy’s lifetime cost.

PROPOSITION 3—Finite stochastic-program policy certificate: *Let $C(u)$ be the convex cost of a nonanticipative finite scenario-tree control problem on a compact convex feasible set $\mathcal{U}$. Let $\hat{u}$ be a feasible reference point, $G(\hat{u})$ its gap defined as in (15), and u^π the feasible tree induced by an evaluated feedback policy. Then*

$$C(u^\pi) - C(\hat{u}) \leq C(u^\pi) - C^* \leq C(u^\pi) - C(\hat{u}) + G(\hat{u}). \quad (16)$$

This certificate is independent of the training residual. It applies at each initial state whose complete scenario tree is evaluated. It is not a bound for an untested continuum of initial conditions, nor does it cover a different shock distribution. The supplement gives the projected-simplex implementation and the independent tree gradient.

3.3. Stationarity and Nonsmooth Solutions

A smooth Hamiltonian can have a strictly suboptimal attracting stationary point. On $[-2, 2]$, consider

$$H(a) = -(a^2 - 1)^2 + 0.2a. \quad (17)$$

Its negative local maximizer is approximately -0.973994 , while its global maximizer is approximately 1.024120 . The former is locally attracting under gradient ascent and has an improvement gap of approximately 0.399875 . Exact evaluation of $V^a(t) = (T - t)H(a)$ does not change that fact. Isolation of stationary points cannot turn an actor-ODE limit into an optimal policy. An optimality conclusion needs, for example, a justified global concavity or gradient-dominance condition together with the necessary representation and evaluation assumptions. Constant-step Adam and finite-step L-BFGS are not covered by such a conclusion merely because they update separate parameter blocks.

Nor does a small differential residual select a viscosity solution. For the exit equation $1 - |v'| = 0$ on $(-1, 1)$ with zero boundary values, the relevant value is $|x| - 1$. The functions

$$w_\varepsilon(x) = \sqrt{1 + \varepsilon^2} - \sqrt{x^2 + \varepsilon^2} \quad (18)$$

have the same boundary values and squared residual tending to zero, but converge to the wrong function $1 - |x|$. At zero, that function fails the viscosity subsolution inequality. Our regression suite retains this example and the biased composite-loss example in Supplement S.2. A convergent approximation of a nonsmooth control problem needs an appropriate mechanism, such as monotonicity, stability, consistency, boundary treatment, and comparison (Barles and Souganidis, 1991); it is not supplied by smooth activation functions alone.

3.4. The Squared-Trace Objective

For a differential critic, let $A = B^\top \nabla^2 v B$ and let z_k be independent standard Gaussian probes. Conditional on a fixed state, action, and parameter vector, put $q_k = z_k^\top A z_k$,

$\bar{q} = K^{-1} \sum_k q_k$, and $r_k = a - q_k/2$. Then

$$\mathbb{E}(a - \bar{q}/2)^2 = (a - \text{tr}(A)/2)^2 + \frac{\|A\|_F^2}{2K}. \quad (19)$$

Averaging r_k^2 instead has an excess expectation independent of K . These are different objectives, even though each individual trace estimate is unbiased.

PROPOSITION 4—An unbiased squared-residual estimand: For $K \geq 2$, define

$$U_K = \frac{(\sum_{k=1}^K r_k)^2 - \sum_{k=1}^K r_k^2}{K(K-1)}. \quad (20)$$

Then $\mathbb{E}U_K = (a - \text{tr}(A)/2)^2$. If differentiation and expectation can be interchanged, its parameter gradient is unbiased for the gradient of that squared residual.

The estimator can be negative in a batch. Unbiasedness does not establish low variance, stability of its minimization, or an accurate policy. Optimizing over actions using the same noisy probes also introduces a selection issue beyond the fixed-action identity. We therefore retain exact derivatives for the differential benchmark checks and assess the probe estimands with repeated independent batches. The executed high-dimensional resource solver uses a finite-step operator and a convex critic; its results are not attributed to Hutchinson acceleration.

4. ENDOGENOUS PREFERENCES AND PORTFOLIO CHOICE

4.1. A Stopped Operating Economy

We first state the continuous-state diffusion benchmark, including its boundary contract and control margins. The executed regional decision experiments use its separately specified, eight-date lottery-settlement counterpart in Section 8. A certificate for that counterpart is an economic statement about its declared transition protocol; transferring it to this diffusion additionally requires the target-error account developed below. We now specify an economy in which the preference-adjustment margin is active in the transition law. Wealth is X , risk aversion is u , and the controls are consumption c , preference adjustment θ , and risky-asset share π . Inside $D = (1.2, 2.8) \times (0.5, 2.0)$, the state evolves as

$$du_t = \theta_t dt + \sigma_u dZ_t^u, \quad (21)$$

$$dX_t = \{(r + \pi_t(\mu - r))X_t - c_t\} dt + \pi_t X_t \sigma_X dZ_t^X, \quad d\langle Z^u, Z^X \rangle_t = \varrho dt. \quad (22)$$

The operating contract terminates at $\tau = T \wedge \inf\{t : S_t \notin D\}$. At that time the agent receives the cardinal payoff

$$G(u, X) = -0.02(u - 2)^2 + 0.1 \log X. \quad (23)$$

The payoff summarizes liquidation or the outside option after leaving the operating regime; it is not a transfer restoring the state to the interior.

The feasible box is $c \in [0.05, 0.8]$, $\theta \in [-0.2, 0.2]$, and $\pi \in [-0.5, 0.8]$. We set

$$T = 1, \quad \rho = 0.04, \quad r = 0.02, \quad \mu = 0.08, \quad \sigma_X = 0.2, \quad \sigma_u = 0.05, \quad \varrho = -0.25. \quad (24)$$

The baseline adjustment-cost coefficient is $k = 2$. For a feasible policy,

$$J_k^\pi(t, u, X) = \mathbb{E}_{t, u, X}^\pi \left[\int_t^\tau e^{-\rho(s-t)} \left\{ \frac{c_s^{1-u_s}}{1-u_s} - \frac{k}{2} \theta_s^2 \right\} ds + e^{-\rho(\tau-t)} G(u_\tau, X_\tau) \right]. \quad (25)$$

Consumption is measured in the fixed units appearing in (25). An additive or multiplicative normalization depending on u changes incentives once u is controlled. We therefore do not treat such transformations as innocuous changes of utility representation.

This boundary contract is a substantive specification. Constant preference volatility cannot remain in a closed interval without an exit or regulator mechanism. Moreover, under the displayed consumption bounds, the drift at $X = 0.5$ is negative even at the largest feasible portfolio share: its maximum is -0.016 . An unregulated state-constrained interpretation is consequently unavailable. Stopping repairs that problem without modifying the interior diffusion, imposing an unreported reflection, or injecting wealth. It is not equivalent to the earlier clipped transition.

The stopped problem has bounded payoffs and Lipschitz coefficients before exit. A useful check is that G is a strict supersolution for every feasible control. The bounds in (24) imply

$$\frac{c^{1-u}}{1-u} - \frac{k\theta^2}{2} + \mathcal{L}^{c, \theta, \pi} G - \rho G \leq -0.8162 \quad (k \geq 0). \quad (26)$$

The supplement derives the bound without a grid maximization. Together with a fixed nondegenerate admissible diffusion, it supplies a boundary-continuity argument for the stopped value. The comparison also has economic content: continued operation has a cost relative to the liquidation payoff. Exit incentives are therefore part of this calibration and must be reported, rather than interpreted as a generic wealth or risk-aversion effect.

4.2. The Operator and Control Margins

Writing subscripts for derivatives, the HJB equation in the interior is

$$\begin{aligned} -V_t = \sup_{c, \theta, \pi} \left\{ \frac{c^{1-u}}{1-u} - \frac{1}{2} k \theta^2 - \rho V + \theta V_u + [(r + \pi(\mu - r))X - c] V_X \right. \\ \left. + \frac{1}{2} \sigma_u^2 V_{uu} + \frac{1}{2} \pi^2 X^2 \sigma_X^2 V_{XX} + \varrho \sigma_u \sigma_X \pi X V_{uX} \right\}, \end{aligned} \quad (27)$$

with $V = G$ on the spatial boundary and at T . Both diagonal diffusion terms and the mixed term belong to the operator. In particular, the generator of u^2 contains σ_u^2 even if the preference drift is zero.

For an interior optimum, the first-order conditions are

$$c^{-u} = V_X, \quad k\theta = V_u, \quad 0 = (\mu - r)XV_X + \pi X^2 \sigma_X^2 V_{XX} + \varrho \sigma_u \sigma_X X V_{uX}. \quad (28)$$

When $V_X > 0$, $V_{XX} < 0$, and all relevant controls are interior, these give

$$c = V_X^{-1/u}, \quad \theta = V_u/k, \quad \pi = -\frac{(\mu - r)V_X}{X\sigma_X^2 V_{XX}} - \frac{\varrho \sigma_u V_{uX}}{X\sigma_X V_{XX}}. \quad (29)$$

The final expression separates a mean–variance term from a preference-hedging term. Its sign is not determined by $\varrho < 0$ alone: the cross derivative and curvature also matter. Nor need the stopped economy satisfy $V_X > 0$ throughout the domain. The constrained implementation compares feasible controls rather than mechanically applying these formulas where their premises fail. At a binding box constraint, the appropriate one-sided derivative or complementary-slackness condition replaces an interior equality.

The cardinal utility derivative is

$$\partial_u \left(\frac{c^{1-u}}{1-u} \right) = \frac{c^{1-u} \{1 - (1-u) \log c\}}{(1-u)^2}. \quad (30)$$

It need not have the sign suggested by a verbal interpretation of increased risk aversion. The full shadow value V_u incorporates this derivative, future opportunities, adjustment costs, covariance, and liquidation. The numerical policy is informative only relative to these declared primitives.

4.3. A Comparative Static That Survives Endogenous Policies

Fix an initial state and let

$$C^\pi = \mathbb{E}^\pi \int_0^\tau e^{-\rho t} \frac{\theta_t^2}{2} dt, \quad B^\pi = J_k^\pi + kC^\pi. \quad (31)$$

For a fixed policy, B^π and C^π are independent of k : changing the coefficient does not change the state law unless the policy is changed. In particular, the stopping time is allowed to be endogenous to the policy.

THEOREM 5—Adjustment costs and cumulative effort: *Suppose the feasible policy class and all primitives other than k are fixed, and optimizers exist at the coefficients being compared. Then $V(k) = \sup_\pi \{B^\pi - kC^\pi\}$ is decreasing and convex. If $k_2 > k_1$ and π_i is optimal at k_i , then*

$$C^{\pi_2} \leq C^{\pi_1}. \quad (32)$$

Each $-C^{\pi_i}$ is a subgradient of V at k_i . At differentiability points, all optimal effort levels equal $-V'(k)$. For policies with objective losses at most e_1, e_2 at the respective coefficients,

$$C^{\pi_2} - C^{\pi_1} \leq \frac{e_1 + e_2}{k_2 - k_1}. \quad (33)$$

The proof uses the two optimality inequalities and is given in Supplement S.3. The approximate version is useful when numerical policies are used in an economic comparison: the effort ordering can be assessed against a welfare-error budget. The result does not hold pointwise for $\theta(t, u, X)$ merely because (29) contains $1/k$. The shadow value, portfolio choice, consumption, and exit distribution all change when the problem is re-solved.

The theorem is an economic comparative static, not an identification result. Inferring k from data additionally requires an observation model for latent preferences, an identified moment map, and control of numerical error in that map. The solver supplies policy and value objects for such a model; a graph of the policy alone cannot establish identification. This distinction leaves the structural-estimation motivation intact while identifying the additional objects it would require.

5. CONTRACT CHANGES AND CERTIFIED POLICY REUSE

5.1. Operating Benefits and the Outside Option

The boundary contract permits a precise economic question: how do portfolio and preference choices respond when the benefit of remaining in the operating regime changes relative to the outside option? Add a constant flow benefit m to the running objective and a constant ℓ to the liquidation payoff, holding the diffusion, feasible controls, stopping rule, and consumption units fixed. These are changes in contract primitives. They need not be changes in the representation of a fixed preference ordering.

For a policy p , let B_n^p be its expected baseline utility, including G but excluding the adjustment charge. In addition to its discounted effort C_n^p , define

$$A_n^p = \mathbb{E}^p \int_{t_n}^{\tau} e^{-\rho(v-t_n)} dv, \quad Z_n^p = \mathbb{E}^p e^{-\rho(\tau-t_n)}. \quad (34)$$

All four functions depend on the initial state. The identity $\rho A_n^p + Z_n^p = 1$ holds pathwise before taking expectations. Thus the same policy's payoff is

$$J_n^p(m, k, \ell) = B_n^p + mA_n^p - kC_n^p + \ell Z_n^p = \ell + B_n^p + dA_n^p - kC_n^p, \quad d = m - \rho\ell. \quad (35)$$

The interpretation of d is the flow benefit of operation net of the annuity value of the liquidation increment. This is a comparison of two explicitly priced components of the contract. It does not require a preference-dependent normalization of consumption utility.

PROPOSITION 6—Contract contrast and economic moments: *Suppose the feasible policy class, transition law, and stopping rule do not depend directly on (m, k, ℓ) , and the required values are finite. Then*

$$V_n(m, k, \ell) = \ell + W_n(d, k), \quad W_n(d, k) = \sup_p \{B_n^p + dA_n^p - kC_n^p\}. \quad (36)$$

A joint change $(m, \ell) \mapsto (m + \rho h, \ell + h)$ leaves the entire optimal-policy correspondence unchanged and adds h to value. The function W_n is convex in (d, k) , increasing in d , and decreasing in k . If p_i has value loss at most e_i at $q_i = (d_i, k_i)$, then

$$(d_2 - d_1)(A_n^{p_2} - A_n^{p_1}) - (k_2 - k_1)(C_n^{p_2} - C_n^{p_1}) \geq -(e_1 + e_2). \quad (37)$$

At points of differentiability, optimal duration and effort equal $\partial_d W_n$ and $-\partial_k W_n$, respectively.

PROOF: Equation (35) proves the first claim for each policy before optimization. The supremum of the displayed affine functions is convex; their slopes in d and k have the stated signs. Approximate optimality gives $J^{p_1}(q_1) \geq J^{p_2}(q_1) - e_1$ and $J^{p_2}(q_2) \geq J^{p_1}(q_2) - e_2$. Adding and cancelling the intercepts proves (37). The differentiability statement follows from the supporting affine function of any optimizer. Q.E.D.

Two implications sharpen the interpretation of the portfolio exercise. First, increasing running utility alone is not a harmless normalization: it changes d and therefore the reward for discounted operating duration. Increasing the terminal payoff by the corresponding annuity value leaves d unchanged and cannot cause a portfolio reversal. Second, observations of choices and state paths alone cannot separately identify m and

ℓ along this equivalence direction. This is an exact restriction on the optimal-policy correspondence, not a claim that d itself is identified. Identification of d still requires an observation model and economically informative variation. If the distribution of the termination time were independent of the policy, A^p would be common to all policies and even d would not change their rankings. Policy-dependent termination is therefore essential to the operating-benefit mechanism.

The supersolution argument for G in the baseline economy is not imposed on every changed contract: adding (m, ℓ) changes its flow-generator comparison by d . The identities above concern payoff functionals and do not require a classical boundary trace for every counterfactual.

The coefficient-weighted charge is kC^p , whereas C^p is an unweighted quantity of adjustment effort. Neither Proposition 6 nor Theorem 5 makes kC^{p_k} monotone. We report both objects. We also distinguish a change in d from a change in correlation: the former changes the relative reward for remaining in operation, while the latter changes the state covariance and need not preserve the same policy-value features.

5.2. Value Envelopes on a Contract Interval

We next turn the economic decomposition into a computable decision rule. Consider a specified finite-horizon Markov model with positive continuation kernels. Its reward, including stopped payments, is affine in $q = (d, k)$; its action sets and transition kernels are fixed across q . A stored policy p_i has independently evaluated features (B^i, A^i, C^i) and hence value

$$J_n^i(s; q) = B_n^i(s) + dA_n^i(s) - kC_n^i(s). \quad (38)$$

Neural actors can supply such policies, but their trained critics are not substituted for these evaluated features.

Let $q_1, \dots, q_M$ be anchor contracts. Suppose $\bar{V}_n^i$ bounds the optimal value above at q_i . For $q = \sum_i \lambda_i q_i$, $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, define

$$L_n(s; q) = \max_i J_n^i(s; q), \quad U_n(s; q) = \sum_i \lambda_i \bar{V}_n^i(s). \quad (39)$$

The weights may be selected to minimize the upper bound; in the one-dimensional contract experiment, adjacent anchors give a chord. The stored policy need not be optimal at its anchor for the lower bound to be valid. The upper bound, in contrast, requires genuine anchor information: an evaluated feasible policy by itself is a lower bound, not an upper one.

THEOREM 7—Uniformly certified policy reuse: *Under the preceding hypotheses, $L_n(s; q) \leq V_n^*(s; q) \leq U_n(s; q)$. Define a feasible switching policy by*

$$i_n(s; q) \in \arg \max_i J_n^i(s; q), \quad p_n^{\text{sw}}(s; q) = p_{i_n(s; q), n}(s). \quad (40)$$

Then, at every date and state,

$$0 \leq V_n^*(s; q) - J_n^{\text{sw}}(s; q) \leq U_n(s; q) - L_n(s; q). \quad (41)$$

If the right-hand side is at most ε on a parameter set, the same policy-construction rule is ε -optimal throughout that set. There is no additional horizon multiplier on this independently evaluated value envelope.

PROOF: Every J^i is feasible, giving the lower bound. For any fixed policy p , affinity gives $J^p(q) = \sum_i \lambda_i J^p(q_i) \leq \sum_i \lambda_i \bar{V}^i = U(q)$. Taking the supremum proves the upper bound. For switching, the terminal equality is common to all stored policies. If $J_{n+1}^{\text{sw}} \geq L_{n+1}$, positivity of the continuation kernel implies

$$J_n^{\text{sw}} = R_n^{p_{in}} + K_n^{p_{in}} J_{n+1}^{\text{sw}} \geq R_n^{p_{in}} + K_n^{p_{in}} J_{n+1}^{in} = J_n^{in} = L_n.$$

Backward induction and the upper bound finish the proof.

Q.E.D.

The feasible-value maximum and switching arguments are established ideas in policy reuse; see [Barreto et al. \(2017\)](#), [Chang \(2021\)](#). The upper envelope is a convex-combination specialization of the policy-cache bound of [Nemecsek and Parr \(2021\)](#). Our use of these results is explicit: the stopped-contract identity identifies the economically relevant parameter, the interval construction below certifies every parameter rather than selected queries, and the action-separation result translates value information into portfolio conclusions. Neither successor features nor policy caching is claimed as a new neural architecture.

The approximation theorem in Section 3 supplies one route to the required anchor upper bounds. If a feasible anchor policy has certified loss e_i , then $J^i(q_i) + e_i$ is an admissible $\bar{V}^i$. For the finite model executed here, backward maximization over an explicit action menu provides the anchor values and an independent feature recursion checks them. The target is this declared finite model: its operator discrepancy is zero relative to itself, not relative to Brownian first passage or to a continuum of actions. The latter approximation questions remain separate.

For a scalar interval $[a, b]$ at fixed k , write the two endpoint-policy lines as $b_a(s) + da_a(s)$ and $b_b(s) + db_b(s)$. Their maximum is a feasible lower bound. The upper chord minus this maximum is concave and piecewise affine in d . Its maximum occurs at an endpoint or at the intersection

$$d_{\times}(s) = \frac{b_b(s) - b_a(s)}{a_a(s) - a_b(s)}, \quad (42)$$

when that intersection is defined and lies in $[a, b]$. The program checks these locations at every date and state. It inserts an anchor at a worst certified parameter and repeats until the maximum gap is below the stated tolerance. A denominator close to zero is handled as two effectively parallel lines, with an explicit numerical tolerance. This is a finite calculation of a continuum bound, not a dense parameter grid relabeled as uniform coverage. Supplement S.7 gives the error version and the numerical slack convention.

5.3. Certifying Economic Decisions, Not Just Values

A welfare bound need not determine a portfolio sign when two actions have nearly equal values. We therefore use a separate separation calculation. Partition the feasible first actions into a desired set A^+ and its complement A^- . For example, A^+ may contain strictly positive portfolio shares and A^- all nonpositive shares. Fix one action $a \in A^+$ and one stored continuation policy p_i . Its one-step value followed by that policy is affine in d :

$$Q_n^i(s, a; d) = R_n(s, a; d) + K_n^a J_{n+1}^i(d). \quad (43)$$

Let $\bar{Q}_n(s, b; d) = R_n(s, b; d) + K_n^b U_{n+1}(d)$, using the same upper chord throughout a contract interval.

PROPOSITION 8—Action separation over an interval: *If the same pair (a, p_i) satisfies*

$$\min_{d \in \{a_0, b_0\}} \left\{ Q_n^i(s, a; d) - \sup_{b \in A^-} \bar{Q}_n(s, b; d) \right\} > 0, \quad (44)$$

then every optimal first action belongs to A^+ for every $d \in [a_0, b_0]$.

PROOF: For every fixed $b \in A^-$, the difference between the feasible tail value and its upper continuation bound is affine on the interval. Positivity at both endpoints therefore implies positivity throughout. Taking the supremum over A^- preserves the strict separation, while $Q^i(s, a; d) \leq Q^*(s, a; d)$ and $Q^*(s, b; d) \leq \bar{Q}(s, b; d)$. No action in A^- can maximize the true finite-model criterion. *Q.E.D.*

The same desired action and continuation policy must be used at both endpoints of a test; selecting unrelated favorable endpoint actions would not prove (44). The implementation enumerates the feasible pairs and records the maximizing pair, both endpoint comparisons, and the resulting interval margin. An unresolved interval is distinguished from a zero effect. This permits a sharp portfolio conclusion away from a switching region while retaining an explicit uncertainty region where the sign is not separated.

5.4. The Exact Scalar-OLS Correspondence

The adaptive reward construction belongs to optimistic linear support (Alegre, Bazzan, and da Silva, 2022). The following observation identifies its exact relation to the adjacent-policy implementation and prevents an implementation shortcut from being mistaken for a new policy-set principle.

PROPOSITION 9—Adjacent policies and the full policy envelope: *Fix k , and suppose each distinct scalar anchor d_i stores one policy that is optimal at every state and date at d_i . If $a < b$ are adjacent anchors, then, pointwise in state and date,*

$$\max_i J^i(d) = \max\{J^a(d), J^b(d)\}, \quad d \in [a, b].$$

Consequently the full policy-line envelope and the adjacent-policy calculation have the same worst chord gap, including its possible corner location.

PROOF: Consider an earlier anchor $d_i < a$. Optimality implies $J^i(d_i) - J^a(d_i) \geq 0$ and $J^i(a) - J^a(a) \leq 0$. This difference is affine, so its slope is nonpositive and $J^i(d) \leq J^a(d)$ for all $d \geq a$. A later anchor $d_i > b$ is similarly dominated by J^b for $d \leq b$. There are no other interior anchors. Taking the maximum proves the assertion. *Q.E.D.*

Dates can be incorporated in a finite state description, so simultaneous statewise coverage is not an obstruction beyond the reach of scalar OLS. It is nevertheless a different numerical objective from coverage at an initial distribution. We compare both objectives using the same exact full-menu oracle and the same 10^{-3} target. The

independent full-envelope calculation verifies Proposition 9; it is not reported as a separate trained implementation of SFOLS. Approximate anchor policies need not obey the adjacent-policy identity, in which case one must retain all feasible policy values or bound the omitted lines.

6. POLICY REUSE WHEN TRANSITION LAWS CHANGE

Reward transfer holds the economic law of motion fixed. Changes in the frequency of a financial regime, the probability of a demand shock, or the survival of installed capacity do not satisfy this restriction. Even with a fixed policy, their repeated effects on future states need not be affine. This section supplies an upper correction for that nonlinearity and a feasible-policy representation with which to certify the resulting counterfactuals.

6.1. The Obstruction and an Endpoint Correction

Consider the finite-horizon additive problem, with a common finite state space and common feasible actions. At date n , write

$$T_{n,\lambda}^a v = R_{n,\lambda}^a + K_{n,\lambda}^a v, \quad (R_{n,\lambda}^a, K_{n,\lambda}^a) = (1 - \lambda)(R_{n,0}^a, K_{n,0}^a) + \lambda(R_{n,1}^a, K_{n,1}^a). \quad (45)$$

Here K is a positive discounted, possibly killed kernel with row mass at most $\gamma_n \leq 1$. Exit payments are included in R ; the terminal function g is common. The parameter λ is known when the policy is selected and is held fixed across dates. Equation (45) can be interpreted as an independent regime draw at each date, not as advance information about future draws.

In contrast to reward-affine transfer, $\lambda \mapsto V_{n,\lambda}$ need not be convex. A two-date uncontrolled chain that pays one only when the first regime is 1 and the second is 0 has value $\lambda(1 - \lambda)$. Both endpoint values are zero and the midpoint value is $1/4$. Interpolating the endpoint optima therefore gives an invalid upper bound. The difficulty remains without a neural approximation, a numerical error, or policy switching.

Fix an interval $[a, b] \subseteq [0, 1]$, put $t = (\lambda - a)/(b - a)$, and let v^a, v^b be endpoint super-solutions:

$$v_n^j \geq \max_u T_{n,j}^u v_{n+1}^j, \quad v_N^j = g, \quad j \in \{a, b\}. \quad (46)$$

Exact endpoint optima are one admissible choice. Define, at every state and action,

$$D_n^u = (K_{n,a}^u - K_{n,b}^u)(v_{n+1}^b - v_{n+1}^a), \quad (47)$$

$$M_N = 0, \quad (48)$$

$$M_n(s) = \max \left\{ 0, \max_{u \in A_n(s)} [D_n^u(s) + \max\{(K_{n,a}^u M_{n+1})(s), (K_{n,b}^u M_{n+1})(s)\}] \right\}. \quad (49)$$

The inner maximum over the two endpoints is taken for the same action u . It does not permit a different control after observing the current regime.

THEOREM 10—A corrected transition-law chord: Under (45)–(46),

$$U_n(\lambda) = (1 - t)v_n^a + tv_n^b + t(1 - t)M_n \quad (50)$$

is an upper bound on $V_{n,\lambda}$ simultaneously at every state, every date, and every $\lambda \in [a, b]$. If $\|K_{n,a}^u - K_{n,b}^u\|_{\infty \rightarrow \infty} \leq L_{K,n}|b - a|$ and $\|v_{n+1}^b - v_{n+1}^a\|_{\infty} \leq L_{v,n+1}|b - a|$, then

$$\|M_n\|_{\infty} \leq (b - a)^2 \sum_{j=n}^{N-1} \Gamma_{n,j} L_{K,j} L_{v,j+1}, \quad \Gamma_{n,j} = \prod_{r=n}^{j-1} \gamma_r. \quad (51)$$

PROOF: Let $C_n = (1 - t)v_n^a + tv_n^b$. Affinity of rewards and kernels gives the exact identity

$$T_{n,\lambda}^u C_{n+1} - C_n = (1 - t)(T_{n,a}^u v_{n+1}^a - v_n^a) + t(T_{n,b}^u v_{n+1}^b - v_n^b) + t(1 - t)D_n^u.$$

The first two terms are nonpositive. Positivity and affinity of K imply $K_{n,\lambda}^u M_{n+1} \leq \max\{K_{n,a}^u M_{n+1}, K_{n,b}^u M_{n+1}\}$ pointwise. Consequently $T_{n,\lambda}^u U_{n+1} \leq U_n$ for every feasible u . Since $U_N = g$, backward comparison proves the first assertion. Taking sup norms in (49) gives $\|M_n\|_{\infty} \leq L_{K,n} L_{v,n+1} (b - a)^2 + \gamma_n \|M_{n+1}\|_{\infty}$; backward iteration gives (51). *Q.E.D.*

The correction depends on the product of a change in the transition operator and a change in continuation value. It vanishes when the kernels are fixed, recovering the reward-transfer upper chord. It also vanishes when the relevant continuation functions do not change. These are economically distinct invariances. Equation (51) concerns the additional upper correction, not the total loss of a policy library: a change in the optimal action can still generate a first-order bank-coverage error. No general square-root oracle-complexity conclusion follows from this equation alone.

An endpoint neural approximation can be used without treating it as an exact optimum. Suppose $\hat{v}_N = g$ and certified one-sided residual bounds satisfy $\max_u T_{n,j}^u \hat{v}_{n+1} - \hat{v}_n \leq r_n^j$. Set $e_N^j = 0$ and $e_n^j = (r_n^j)_+ + \gamma_n e_{n+1}^j$. Then $v_n^j = \hat{v}_n + e_n^j$ satisfies (46), by positivity and the row-mass bound. Thus the theorem accommodates approximate neural endpoints and an explicit evaluation budget. A sampled training residual does not supply the required r_n^j . The numerical implementation below uses exact finite-model endpoints so that the transition-law correction, rather than endpoint estimation, is isolated.

6.2. Feasible Policy Values over the Entire Parameter Interval

Fix a deterministic Markov policy p that does not observe future regimes. With $h = N - n$, its value is a polynomial of degree at most h :

$$J_n^p(\lambda) = \sum_{j=0}^h \binom{h}{j} \lambda^j (1 - \lambda)^{h-j} F_{n,j}^p. \quad (52)$$

The vector coefficients have a direct conditional-expectation interpretation. Imagine drawing all h future regime labels before the experiment, but not revealing them to p . Conditional on exactly j labels being 1, the current label is 1 with probability j/h . Hence $F_{N,0}^p = g$ and

$$F_{n,j}^p = \left(1 - \frac{j}{h}\right) \left[R_{n,0}^{p_n} + K_{n,0}^{p_n} F_{n+1,j}^p\right] + \frac{j}{h} \left[R_{n,1}^{p_n} + K_{n,1}^{p_n} F_{n+1,j-1}^p\right]. \quad (53)$$

A term with zero weight is omitted, so no nonexistent coefficient is used at $j = 0$ or $j = h$. The recursion remains valid with killing: labels after exit can be drawn without changing the stopped payoff.

The representation permits an operational continuum certificate. Restrict the polynomial coefficients to a closed parameter cell by de Casteljau subdivision. Express (50) at the same degree by degree elevation. For a library $\mathcal{P}$ and any cell I , let $U_{n,j}^I$ and $F_{n,j}^{p,I}$ denote these coefficients and put

$$\epsilon_I = \max_{n,s} \left[\min_{p \in \mathcal{P}} \max_{0 \leq j \leq N-n} \{U_{n,j}^I(s) - F_{n,j}^{p,I}(s)\} \right]_+. \quad (54)$$

The order of the minimum and maximum deliberately gives an upper bound rather than an equality.

COROLLARY 11—Transition-law policy reuse: *Choose at each state and date an index attaining $\max_{p \in \mathcal{P}} J_n^p(\lambda)$ and use that policy's current action. The resulting feasible switching policy p_λ^{sw} satisfies*

$$0 \leq V_{n,\lambda}(s) - J_n^{p_\lambda^{\text{sw}}}(s) \leq \epsilon_I, \quad \lambda \in I. \quad (55)$$

The maximum of ϵ_I over a partition therefore certifies every parameter in the partition, every state, and every date, without interpreting parameter samples as a uniform test.

PROOF: The Bernstein basis is nonnegative and sums to one. For each fixed policy, the polynomial difference on I is bounded above by its largest coefficient difference. Taking a minimum over policies proves $U_n - \max_p J_n^p \leq \epsilon_I$. Backward induction and positivity give $J_n^{p^{\text{sw}}} \geq \max_p J_n^p$: after taking the selected policy's current action, subsequent switching cannot lower its continuation value. Theorem 10 completes the proof. Q.E.D.

The same lower representation can be paired with a different upper oracle. For example, disclose to a relaxed controller the number of remaining regime-1 draws, and disclose each realized label afterward. Replace the fixed action in (53) by a maximum over a *common* current action after combining the two weighted terms. Call the resulting vectors $W_{n,j}$. Then $\sum_j \binom{h}{j} \lambda^j (1-\lambda)^{h-j} W_{n,j}$ is an upper value: every original policy is feasible in the enlarged information structure. This is an explicit, zero-penalty information relaxation in the sense of Brown, Smith, and Sun (2010), not a new general duality principle. Its triangular count recursion requires $h+1$ coefficient vectors at a date with h periods left, rather than enumeration of all 2^h label histories. We report its information premium separately from the corrected-chord certificate; a loose information bound is not evidence of a large policy loss.

Changing transitions has long been a subject of parametric and robust dynamic programming. Our result does not claim that such counterfactuals were previously impossible. Its additional content relative to a reward-policy cache is the explicit cross-operator term in (49), its second-order upper-error account, and its combination with feasible policy polynomials and switching. These objects retain the same scalar parameter across the horizon rather than allowing an adversary to reset it freely at each node. In the application, the certificate supplies a new, fully specified uncertainty counterfactual for the original stopped preference economy.

6.3. The Price of Compressing an Upper Oracle

An upper value can be valid without being the sharpest available upper value. This distinction matters when an interval is to be accepted at a prescribed welfare tolerance.

The relevant comparison holds the interval, feasible policies, and subdivision cells fixed. Let $I = [a, b]$, $t = (\lambda - a) / (b - a)$, and $h = N - n$. A localized count-informed recursion has coefficients

$$W_{n,j}^I = \max_u \left\{ (1 - j/h) T_{n,a}^u W_{n+1,j}^I + (j/h) T_{n,b}^u W_{n+1,j-1}^I \right\}. \quad (56)$$

A term with zero weight is omitted. At the terminal date the coefficient is g . The maximization uses the *same* action in the two branches. Conditional on the number of remaining draws from endpoint law b , a draw is from b with probability j/h . The enlarged controller learns the remaining count, whereas an admissible policy in the original economy can ignore it. Thus the associated polynomial is an upper value, not a deployable anticipative policy.

For comparison, write the degree- h coefficients of the corrected chord as

$$U_{n,j}^I = (1 - j/h) v_n^a + (j/h) v_n^b + f_{h,j} M_n, \quad f_{h,j} = \frac{j(h-j)}{h(h-1)} \quad (h \geq 2). \quad (57)$$

The correction is zero at a one-period horizon. A third upper recursion allows the endpoint law to be chosen afresh at every date and state:

$$B_N^I = g, \quad B_n^I = \max_{u,e \in \{a,b\}} T_{n,e}^u B_{n+1}^I. \quad (58)$$

This last recursion is a finite-horizon, reward-valued rectangular relaxation. It discards the common-parameter restriction more extensively than the count construction.

PROPOSITION 12—Matched upper-oracle ordering: *Under the hypotheses of Theorem 10, $W_{n,j}^I \leq U_{n,j}^I$ for every state, date, and coefficient. Moreover, $W_{n,j}^I \leq B_n^I$. Positive Bernstein subdivision preserves both orderings. With identical feasible-policy coefficients and cells, the localized count certificate is no greater than either comparator's certificate. No ordering between U^I and B^I is required.*

We thank the advisory referee for identifying the first ordering. Its economic-computational implication is a tradeoff between upper precision and the work required to obtain it. The proof, including the endpoint and short-horizon cases, appears in Supplement S.10. A two-period example makes the difference strict: the original value is $\max(t, 1 - t)$, the chord coefficients are $(1, 1, 1)$, and the count coefficients are $(1, 1/2, 1)$. At $t = 1/2$, their upper values are 1 and $3/4$, respectively, against a true value of $1/2$.

The correction compresses the count dependence. Its proof replaces a count-dependent convex combination of the two propagated corrections by their maximum and replaces endpoint action backups by endpoint supersolutions. It thereby trades precision for fewer action-wide kernel applications. With H remaining dates and already available exact endpoint solutions, the implementation requires $4H - 6$ such applications for the correction when $H \geq 2$, versus $H(H + 1) - 2$ for the count recursion with cached neighboring continuations and shared endpoint coefficients. The rectangular recursion requires $2H$. These are operation counts for the specified recursions, not ratios of elapsed times.

The nonzero correction stores $(H - 1)S$ additional values for S states, against $H(H - 1)S/2$ interior count coefficients. Endpoint values are shared. The implementation expands one date's chord coefficients only when a certificate needs them. This saving

does not remove the common feasible-policy bank: m policies require order mSH^2 value coefficients and mSH policy indices. Action-wide temporary arrays and the transition representation also remain. A comparison of total work must therefore include anchor optimization, policy evaluation, local restriction, and subdivision, not only the upper recursion.

Regional bounds have a substantial verification literature. Parameter lifting in [Quatmann, Dehnert, Jansen, Junges, and Katoen \(2016\)](#) relaxes repeated parameter dependence and refines regions; its two-toss example already exhibits the obstruction behind $t(1 - t)$. That work principally treats probabilistic specifications in parametric Markov models, whereas here the target is a finite-horizon reward optimum together with an admissible policy lower value. Equation (58) is our explicitly specified regional comparator, not an implementation or timing claim about their verification software. It allows datewise as well as statewise endpoint resets. The count construction retains a common count of the endpoint draws but reveals that count to the optimizing controller. Fixed-policy Bernstein coefficients retain the original repeated-parameter dependence without that information. The corrected chord retains its signed cross-operator term while compressing the propagation of the correction. These are different relaxations of dependence, not alternative names for the same algorithm.

6.4. Total Error and Finite Refinement

The second-order correction in Theorem 10 is not a second-order bound on the loss of an endpoint-policy bank. A bound on the latter must allow optimal actions to change. For a finite target, let r_n bound the absolute reward, let $\beta_n \leq 1$ bound every kernel row mass, and let

$$\rho_n = \max_u \|R_{n,1}^u - R_{n,0}^u\|_\infty, \quad \kappa_n = \max_u \|K_{n,1}^u - K_{n,0}^u\|_{\infty \rightarrow \infty}.$$

Define nonnegative constants backward by

$$\begin{aligned} V_n &= \|g\|_\infty, & V_n &= r_n + \beta_n V_{n+1}, \\ L_n &= 0, & L_n &= \rho_n + \kappa_n V_{n+1} + \beta_n L_{n+1}, \\ Q_n &= 0, & Q_n &= \kappa_n L_{n+1} + \beta_n Q_{n+1}. \end{aligned} \tag{59}$$

These constants depend on primitives of the stored finite model, not on a smooth optimal-policy correspondence.

THEOREM 13—A total coefficient-error bound: *Suppose the two endpoint policies are optimal at every state and date. On an interval of width w , the corrected-chord certificate with those policies satisfies*

$$\epsilon_I \leq \max_{0 \leq n < N} \{2L_n w + \frac{1}{2} Q_n w^2\}. \tag{60}$$

The same upper estimate applies to the localized count certificate. Put $L_ = \max_n L_n$ and $Q_* = \max_n Q_n$. For $\epsilon > 0$, dyadic refinement terminates once*

$$w \leq \min\{1, \epsilon / (4L_*), \sqrt{\epsilon / Q_*}\},$$

where a quotient by a zero constant imposes no restriction. Thus, for fixed finite-model constants, the worst-case number of scalar intervals is $O(\epsilon^{-1})$; an $O(\epsilon^{-1/2})$ policy-bank claim does not follow from the correction alone.

The proof bounds optimal values and all fixed-policy values by the same Lipschitz recursion, and then bounds their *coefficients*, not just sampled values. It is given in Supplement S.10. The result permits policy kinks. It does not remove dependence on the state/action target or furnish a high-dimensional parameter covering rate. Its practical use is to establish termination while allowing the realized coefficient certificate to be much tighter than a global Lipschitz estimate.

A useful allocation rule follows from Proposition 12. On each dyadic interval, first compute the compressed certificate. If it fails the target, compute the local count certificate before buying another anchor. Split only when that second certificate fails. For exact endpoint oracles and the same local endpoint-policy rule, this cascade never requests an anchor outside the tree requested by chord-only refinement: every interval accepted by the chord is also accepted by count, and every failed count interval would have failed the chord. Extra count work can reduce anchor demand, but need not reduce elapsed time. Our matched frontier below compares each upper method on common banks. The separately executed autonomous experiment reports chord, count, and cascade refinement from fresh policy caches; its timing is not inferred from the common-bank frontier.

7. PREFERENCE FORMATION AND THE BOUNDARY BETWEEN RISK CHOICES

The annuity identity identifies which contract changes matter; it does not identify the source of a portfolio reversal. To distinguish an operating-duration incentive from preference formation, fix an initial state and date and partition admissible policies by their *first* risky position. Let $\mathcal{P}_+$ require a positive first position and $\mathcal{P}_-$ a nonpositive one. Later actions remain unrestricted within the economic model. In the finite implementation the first position is held over the first rebalancing period, of length $\delta = 1/8$, until liquidation. These are nonempty finite policy sets. A continuous-time counterpart must restrict the position over a positive-duration interval, not merely its value at the initial instant; Section 8 makes this distinction explicit. For that counterpart, replace maxima by suprema unless attainment is established. The finite-policy conclusions below concern the declared calendar.

Write

$$W_\sigma(d, k) = \max_{p \in \mathcal{P}_\sigma} \{B_p + dA_p - kC_p\}, \quad \Delta(d, k) = W_+(d, k) - W_-(d, k), \quad \sigma \in \{+, -\}. \quad (61)$$

Thus $\Delta > 0$ identifies a strictly positive optimal first position, while $\Delta < 0$ rules it out. A zero is an indifference, not necessarily a unique threshold. The economic objects A_p and C_p are discounted duration and adjustment effort, not the first-period duration or the charge kC_p .

PROPOSITION 14—Duration, adjustment effort, and a risk-switching locus: *For finite policy sets, W_+ and W_- are convex piecewise-affine functions of (d, k) . At a point where the optimizing feature vector in each class is locally constant,*

$$\Delta_d = A_+ - A_-, \quad \Delta_k = -(C_+ - C_-). \quad (62)$$

If $\Delta(d^, k) = 0$ and $A_+ \neq A_-$ at such a point, the local switching locus has slope*

$$\frac{dd^*(k)}{dk} = \frac{C_+ - C_-}{A_+ - A_-}. \quad (63)$$

More generally, on an interval of d at a fixed k , suppose every pair of optimizing policies, one from each class, satisfies $A_+ - A_- \geq \kappa > 0$. Then Δ is strictly increasing with slope at least κ in the integral sense. Opposite endpoint signs imply exactly one crossing. If a computed difference has uniform error at most $e_+ + e_-$, any zero of that computed difference lies within $(e_+ + e_-)/\kappa$ of the true crossing.

PROOF: A maximum of finitely many affine functions is convex, Lipschitz on compact sets, and piecewise affine. On a cell with fixed optimizing feature vectors, differentiation gives (62); implicit differentiation gives (63). Across cells the value functions are absolutely continuous. Their derivatives almost everywhere are the features of active optimizers, a finite-policy instance of the envelope argument in Milgrom and Segal (2002). Integrating the assumed derivative separation gives $\Delta(d_2, k) - \Delta(d_1, k) \geq \kappa(d_2 - d_1)$. Continuity gives existence and the strict inequality gives uniqueness. Finally, at an approximate zero the absolute true difference is at most $e_+ + e_-$; the same slope inequality bounds its distance from the true zero. Q.E.D.

The slope ratio gives a contract-equivalent price of differential preference adjustment. If positive-risk policies both keep the agent in operation longer and use more adjustment effort, a higher adjustment cost requires a larger operating benefit before the positive-risk class becomes preferable. If the effort ordering is reversed, the comparative static reverses. The aggregate theorem that optimal effort decreases with k does not determine this ratio: it compares optimal choices across costs, whereas (63) compares two constrained risk classes at the same cost. The duration-separation hypothesis must be established over the claimed interval; a bisection calculation or a derivative at one point does not establish it.

7.1. A Matched Preference-Formation Counterfactual

Let W_σ^{adj} retain the full adjustment set and let W_σ^0 restrict $\theta = 0$ at every state and date. The state u remains stochastic, with the original volatility, covariance, operating region, and settlement rule. Thus the comparison removes deliberate preference formation, not preference risk, and changes no reward level or financial technology. Define its option value in each risk class by

$$\Omega_\sigma(d, k) = W_\sigma^{\text{adj}}(d, k) - W_\sigma^0(d), \quad \Omega_\sigma \geq 0. \quad (64)$$

The zero-adjustment model does not depend on k because its realized adjustment effort is zero.

PROPOSITION 15—Risk-specific option values: *At every contract,*

$$\Delta^{\text{adj}}(d, k) - \Delta^0(d) = \Omega_+(d, k) - \Omega_-(d, k). \quad (65)$$

In particular, preference formation reverses a nonpositive-risk ranking, $\Delta^0 < 0$, precisely when $\Omega_+ - \Omega_- > -\Delta^0$. It reverses a positive-risk ranking, $\Delta^0 > 0$, precisely when $\Omega_- - \Omega_+ > \Delta^0$. If Δ^0 has the same strict sign at two contracts but Δ^{adj} has opposite signs, the change in relative option values identifies the incremental force behind that reversal, rather than the generic annuity identity alone.

PROOF: Set inclusion gives nonnegativity in (64). Subtract the two definitions of Ω_σ to obtain (65). The two strict inequalities are then necessary and sufficient for opposite signs. No interior first-order condition or portfolio-distance argument is required. *Q.E.D.*

The propositions suggest a focused experiment. We independently optimize both risk classes with and without deliberate adjustment, record (B, A, C) for the attaining policies, and compare their value differences. We also vary k while keeping the operating region, covariance, and settlement curvature fixed. The resulting brackets locate an indifference by continuity and opposite signs, but are not reported as a globally unique switching threshold. The full first-action sign certificates from Section 5 retain their separate, stronger exclusion interpretation where applicable. Utility units are the same fixed cardinal units as in the original stopped economy; the ratio (63) translates adjustment-cost changes into operating-benefit changes without pretending to estimate either from data.

7.2. A Primitive Condition for a Risk-Specific Adjustment Option

The option decomposition identifies the quantity that changes a risk ranking; a primitive condition is needed to predict its sign. Consider a two-stage specialization that isolates the preference channel. A risk class $\sigma \in \{+, -\}$ delivers a positive consumption lottery C_σ and an exogenously specified discounted operating duration A_σ . Before consumption is realized, the agent chooses $\theta \in [-\bar{\theta}, \bar{\theta}]$ at cost $k\theta^2/2$. Set

$$F_\sigma(\theta) = \mathbb{E} \frac{C_\sigma^{1-u_0-\theta}}{1-u_0-\theta}, \quad B_\sigma = F_\sigma(0), \quad g_\sigma = F'_\sigma(0), \quad M_\sigma = \max_{|\theta| \leq \bar{\theta}} [-F''_\sigma(\theta)],$$

where $u_0 > 1 + \bar{\theta}$ and consumption has finite positive support. The class value is $B_\sigma + dA_\sigma + \Omega_\sigma(k)$, with $\Omega_\sigma = \max_{|\theta| \leq \bar{\theta}} \{F_\sigma(\theta) - B_\sigma - k\theta^2/2\}$. Unlike the optimized-policy moments in (62), g_σ and M_σ are computed directly from the consumption lottery and preference index.

PROPOSITION 16—Consumption exposure and the adjustment premium: For $u > 1$,

$$U_{uu}(c, u) = -\frac{c^{1-u}}{(u-1)^3} \{[1 + (u-1) \log c]^2 + 1\} < 0. \quad (66)$$

If $|g_\sigma|/(k + M_\sigma) \leq \bar{\theta}$, then

$$\frac{g_\sigma^2}{2(k + M_\sigma)} \leq \Omega_\sigma(k) \leq \frac{g_\sigma^2}{2k}. \quad (67)$$

Consequently, if $A_+ - A_- =: D_A > 0$ and

$$\underline{\Omega} := \frac{g_+^2}{2(k + M_+)} - \frac{g_-^2}{2k} > 0, \quad (68)$$

preference adjustment lowers the risk-indifference contract by at least $\underline{\Omega}/D_A$. Specifically, with $d^0 = (B_- - B_+)/D_A$, every $d \in (d^0 - \underline{\Omega}/D_A, d^0)$ strictly favors the nonpositive class without adjustment and the positive class with adjustment.

The argument uses global concavity in the preference index, not an interior first-order condition for an optimized portfolio. Low consumption affects the adjustment incentive through $U_u(c, u) = c^{1-u}[1 + (u-1)\log c]/(u-1)^2$. A downside consumption exposure can therefore increase the magnitude of the preference-adjustment benefit even when its unadjusted utility is lower. Condition (68) compares that gain with curvature and effort cost; greater risk by itself is not sufficient. Supplement S.11 proves the bounds and shows how M_σ can be bounded using endpoint preferences rather than an unverified curvature grid.

For a declared family, take $u_0 = 2$, $\bar{\theta} = 0.2$, $C_- = 0.5$, and $C_+ = 0.05$ with probability p and 0.8 otherwise, with $D_A = 0.5$. For every $p \in [0.06, 0.10]$ and $k \in [20, 80]$, the primitive bound gives $\Omega_+ - \Omega_- > 0.00186$ and a threshold reduction exceeding 0.00372 in operating-benefit units. This is a continuum prediction from the primitive inequalities, not interpolation of a nine-point exercise. The finite support and preference interval imply $|g_+| \geq 1.4820$, $|g_-| \leq 0.6138$, and $M_+ \leq 17.342$; these conservative bounds imply the stated uniform result. The larger, point-specific lower bounds and the exactly optimized options are reported in the supplement. This specialization explains a channel; it is not offered as a calibration or approximation of the full stopped diffusion. The full finite economy is examined separately with decision-specific certificates.

7.3. Certifying a Risk Ranking over a Joint Counterfactual Region

A welfare tolerance is not a tolerance for the sign of a small class-value difference. We therefore construct upper and lower values separately for each risk class and each adjustment regime. Fix $I_\lambda = [a, b]$ and $I_d = [d_l, d_h]$ and write $t = (\lambda - a)/(b - a)$ and $s = (d - d_l)/(d_h - d_l)$. For a fixed policy, its value is polynomial in t and affine in d . For a fixed λ , the optimal class value is convex in d , since the feasible policy set is unchanged and every policy value is affine.

Let $U_\sigma^l(t)$ and $U_\sigma^h(t)$ be valid transition-polynomial upper values at the two contracts, constructed using the constrained count or corrected-chord recursion. Then

$$U_\sigma(t, s) = (1 - s)U_\sigma^l(t) + sU_\sigma^h(t)$$

is an upper value throughout the rectangle. Evaluate any feasible class-policy bank independently to obtain lower polynomials $L_{\sigma,p}(t, s)$. Denote their tensor Bernstein coefficients on a cell by $U_{\sigma,ij}$ and $L_{\sigma,p,ij}$.

PROPOSITION 17—Decision-specific tensor certificate: *Suppose the upper and lower polynomial constructions, including their coefficient operations, each have absolute error at most e . Throughout the closed cell,*

$$\begin{aligned} \underline{\Delta} &:= \max_{p \in \mathcal{B}_+} \min_{i,j} \{L_{+,p,ij} - U_{-,ij}\} - 2e \leq \Delta, \\ \Delta &\leq \bar{\Delta} := \min_{p \in \mathcal{B}_-} \max_{i,j} \{U_{+,ij} - L_{-,p,ij}\} + 2e. \end{aligned} \tag{69}$$

In particular, $\underline{\Delta}^{\text{adj}} > 0$ and $\bar{\Delta}^0 < 0$ certify opposite risk rankings under the same contract and shock-law parameter. They also imply $\Omega_+ - \Omega_- \geq \underline{\Delta}^{\text{adj}} - \bar{\Delta}^0 > 0$.

PROOF: A tensor Bernstein polynomial is a convex combination of its coefficients. Each positive-class policy is a feasible lower value and each negative-class upper

polynomial bounds that class optimum; their difference therefore bounds Δ from below. Taking a coefficient minimum, followed by a maximum over feasible policies, preserves that bound. Interchanging the classes gives the upper inequality. The two absolute error allowances add. Applying the option decomposition gives the final statement. *Q.E.D.*

This certificate does not require uniform duration separation, a regular active-policy cell, or uniqueness of a threshold. Its object is a strict choice on a declared region. It uses the same transition parameter in all periods, permits later risky positions to change sign, and imposes the adjustment restriction at every date. In the application below, the class-value tolerances are generated by these coefficient comparisons rather than borrowed from the 10^{-3} generic welfare result.

8. THE ECONOMIC DECISION AND ITS ROBUSTNESS

The object of a portfolio certificate is a decision under a specified contract, not a sign attached to an unspecified limiting model. Three primitives are particularly important here: the duration for which the initial position is chosen, the settlement technology that determines subsequent states, and the cardinal value assigned to a change in preferences. We make these primitives explicit and study their effects without changing the financial controls, adjustment technology, or liquidation rule of the baseline experiment.

8.1. *A Rebalancing Contract with a Finite Settlement Lottery*

The finite economy has eight rebalancing dates over one year. At each date the agent observes (u, X) , chooses (c, θ, π) , and holds that control until the next date or liquidation, whichever occurs first. The preference grid contains 33 tiers on $[1.2, 2.8]$; the baseline wealth grid contains 49 balances on $[0.5, 2]$. The union of the two action meshes has 1,565 elements. Three deposited state- and date-dependent feasible proposals complete the 1,568-action baseline menu. Removing those proposals from a lower-policy construction does not remove them from the target against which that policy is certified.

Here is a primitive description of the transition experiment. Independently at every date, a law indicator chooses the correlation -0.25 with probability $1 - \lambda$ and 0.25 with probability λ . The probability λ is fixed for the entire contract; the indicator is not observed before the action. Conditional on the indicator, two independent symmetric signs determine a proposed increment using the drift and volatility in (21)–(22). For an interval $h = 1/8$ and signs (z_u, z_X) , that increment is

$$h \left((r + \pi(\mu - r))X - c \right) + \sqrt{h} \left(\pi X \sigma_X \{ \varrho z_u + \sqrt{1 - \varrho^2} z_X \} \right). \quad (70)$$

If the line segment from the current state to the proposed state first meets the boundary at fraction $\alpha < 1$, liquidation takes place at time αh , at that boundary point. Otherwise $\alpha = 1$ and the arrival is settled on the four neighboring grid nodes by independent, mean-preserving two-point lotteries in the two coordinates. Flow utility and the operating benefit accrue for αh , with the discount annuity $(1 - e^{-\rho \alpha h})/\rho$; liquidation pays $e^{-\rho \alpha h} G$. The complete conventions, including arrival on a boundary node, are given in Supplement S.12.

This is a discrete rebalancing and randomized-settlement contract. The grid is part of its transition technology: it describes admissible recorded balances and preference tiers, not an assertion that interpolation has no economic effect. It is an analytical economic specification, not a calibration of an observed intermediary or an empirical claim that wealth is randomly rounded. The continuous diffusion in Section 4 remains a separate economic benchmark. Both specifications preserve consumption, costly preference formation, risky investment, stochastic preferences, covariance, stopping, and liquidation. Their relationship is an approximation question rather than an identity.

An interior arrival Y settled to Z satisfies

$$\mathbb{E}[Z | Y] = Y, \quad \text{Cov}(Z) - \text{Cov}(Y) = \text{diag}(\mathbb{E}[(Y_i - l_i(Y))(r_i(Y) - Y_i)]). \quad (71)$$

The quantities l_i, r_i are adjacent grid nodes. Consequently, changing settlement granularity alters risk even when the rebalancing calendar and action menu are held fixed. The wealth-lottery perturbations below exploit this observation: they change an economically relevant primitive and reveal the movement of the original decision region. They are not used to assert an unproved diffusion limit.

8.2. A Positive-Duration Risk Choice

Let $\mathcal{P}_+^\delta$ require the initial risky position to be positive throughout the first commitment interval of length δ , until liquidation, and let $\mathcal{P}_-^\delta$ require it to be nonpositive. In the finite experiment the control is constant within the first interval, so this is precisely the first-action partition used by the solver. Later positions are unrestricted. In continuous time a constant initial position is one possible commitment contract; an adapted sign restriction over the same interval is a different admissible commitment contract. Either can be studied, but they must not be interchanged without changing the policy sets.

PROPOSITION 18—The duration of a risk classification: *For the diffusion in Section 4, changing a bounded measurable control at a single deterministic initial instant leaves its state path and payoff unchanged, up to the usual indistinguishability convention. Hence a partition that restricts only the sign of that isolated control value has equal class suprema. A strictly signed lifetime class-value difference instead concerns a positive-duration restriction such as $\mathcal{P}_\sigma^\delta$. When a time mesh is refined while the latter estimand is held fixed, the restriction must continue to cover the same interval δ .*

PROOF: Changing a bounded drift integrand on a singleton changes no Lebesgue integral. The difference of the corresponding stochastic integrals has quadratic variation equal to the integral of a bounded squared integrand over that singleton, which is zero. The resulting state and stopped payoff therefore coincide. Either initial sign can be assigned without changing the value. Holding the duration of a restriction fixed under refinement is then a condition on the experiment, not an attainment assumption. *Q.E.D.*

The economic horizon of the reported choice is thus one eighth of a year. This is distinct from the lifetime duration feature A_p in (61). A local Hamiltonian comparison has flow-value units and can also be informative, but it is not the lifetime commitment-value difference reported here.

8.3. Preferences over Preference Changes

Consumption risk aversion does not identify the cardinal payoff from changing the preference index. A useful explicit specification is

$$U_b(c, u) = \int_1^c x^{-u} dx + b(u) = \frac{c^{1-u} - 1}{1-u} + b(u). \quad (72)$$

All members have marginal consumption utility c^{-u} and relative risk aversion u . The reference consumption level is one in the fixed units of the economy; $b(u)$ is the payoff from being in preference state u at that reference consumption. It is a preference over preference states. It must be specified or, in an empirical application, separately disciplined. The baseline corresponds to $b_0(u) = 1/(1-u)$. Thus its cross-preference cardinal comparison is a substantive primitive rather than a consequence of the CRRA coefficient alone.

Consider first $b(u) = b_0(u) + a(u - u_0)$ in the two-stage family of Section 7.2. A common tilt a changes neither class's consumption lottery nor its fixed- u_0 payoff. Nevertheless, it changes the shadow benefit of adjustment in both classes. The following uniform result characterizes a nontrivial set on which the original economic mechanism survives.

PROPOSITION 19—A cardinally robust primitive adjustment premium: *Retain $u_0 = 2$, the adjustment bound 0.2, $C_- = 0.5$, $C_+ \in \{0.05, 0.8\}$ with probability $p \in [0.06, 0.10]$ on the lower outcome, and $k \in [20, 80]$. For every common tilt $a \in [-0.25, 0.25]$,*

$$\Omega_+(a) - \Omega_-(a) \geq \frac{(1.482 - 0.25)^2 20 / (20 + 17.342) - (0.6138 + 0.25)^2}{160} > 0.0004173. \quad (73)$$

For a duration difference $D_A = 0.5$, the adjusted indifference contract is therefore at least 0.0008346 below the unadjusted one throughout this probability–cost–cardinal family.

The proof in Supplement S.12 uses the same curvature restriction as the original proposition and treats the common cost in both option bounds jointly. It is an analytic continuum statement, not an inference from a table of optimized examples. Its sufficient tilt radius is conservative. Conversely, at $(p, k) = (0.08, 40)$ the relative option is 0.04882837 at $a = 0$ but -0.01391977 at $a = 1$. The unadjusted indifference is 1.5 in both cases; the adjusted indifference changes from 1.40234327 to 1.52783953. The larger tilt is outside the robust family, and the reversal is retained as economically informative. Downside exposure and the cardinal value of preference changes jointly determine the mechanism.

8.4. A Dynamic Accounting and Reoptimization Experiment

The dynamic economy does not inherit the two-stage direction mechanically. Even without deliberate adjustment, its preference state is stochastic. We therefore recompute both risk classes in both adjustment regimes under each change in primitives. In particular, an additive tilt need not leave the dynamic no-adjustment values unchanged.

For a policy p , let F_p be discounted exposure to $(c^{1-u} - 1)/(1-u)$; let B_p and H_p be the corresponding exposures to $1/(1-u)$ and $u - 2$; and retain discounted operating duration A_p , quadratic adjustment effort C_p , and liquidation payoff G_p . Under a

normalization multiplier s , tilt a , and liquidation multiplier z , the exact policy value is

$$J_p(s, a, d, k, z) = F_p + sB_p + aH_p + dA_p - kC_p + zG_p. \quad (74)$$

The baseline is $(s, a, k, z) = (1, 0, 2, 1)$. All six features use the stopped transition protocol and include its endogenous exit distribution.

PROPOSITION 20—Dynamic feature accounting with reoptimization: *Fix the finite transition law and the class and adjustment restrictions. The optimized value is the maximum of the affine forms in (74). At a differentiability point its derivatives in (s, a, d, k, z) are $(B, H, A, -C, G)$ for an optimizing policy. If p_0 is optimal at $q_0 = (s_0, a_0, d_0, k_0, z_0)$, then for any q_1 ,*

$$W(q_1) - W(q_0) = J_{p_0}(q_1) - J_{p_0}(q_0) + \mathcal{R}(q_1; p_0), \quad \mathcal{R}(q_1; p_0) \geq 0. \quad (75)$$

The identity holds separately for all four class–regime values. For the relative option, subtract the two nonpositive-class and no-adjustment comparisons with their appropriate signs; the resulting difference of reoptimization gains has no predetermined sign.

PROOF: There are finitely many deterministic Markov policies. The affine representation gives the envelope statement. Add and subtract $J_{p_0}(q_1)$, use $W(q_0) = J_{p_0}(q_0)$, and use feasibility of p_0 at q_1 to obtain (75). Taking the four signed differences preserves the identity, but not the sign of its remainder. Q.E.D.

Equation (75) separates the changed valuation of a fixed exposure from endogenous policy adjustment. It does not identify a causal contribution from data or decompose interacting primitives into invariant shares. For changes in the shock law we directly reevaluate the baseline policy under the new law and again report the nonnegative gain from reoptimization. That calculation changes the feature distribution itself, rather than treating covariance as a reward coefficient. We also solve a zero-covariance law: setting $\lambda = 1/2$ would mix two nonzero-correlation shock laws and is not the same intervention.

The results reported below use the original $k = 2$ and all financial and preference controls. They explain why a positive relative adjustment option can coexist with positive risk in both regimes, with nonpositive risk in both regimes, or with opposite choices. That separation is the economic content needed to connect a computed certificate to the mechanism of the full model.

8.5. Decision Error across Economic Targets

Let $[\ell_r, u_r]$ be a certified difference interval for a finite protocol in regime $r \in \{\text{adj}, 0\}$, including its numerical arithmetic allowance. If another target satisfies $|W_\sigma^{r, \text{target}} - W_\sigma^{r, \text{finite}}| \leq \eta_\sigma^r$, then

$$\Delta^{r, \text{target}} \in [\ell_r - \eta_+^r - \eta_-^r, u_r + \eta_+^r + \eta_-^r]. \quad (76)$$

This follows by subtracting the two class-value error intervals. It must not charge an already included arithmetic error a second time. For the historical rectangle, an additional common per-class error strictly below $2.6496827802505486 \times 10^{-5}$ would suffice for both original signs. That number is a required budget, not an estimate of the diffusion error.

Our robustness experiment instead uses explicitly specified finite settlement protocols and reoptimizes each. It retains all four corners and the center of the historical rectangle, reports its failures as well as its successes, brackets the movement of indifference contracts, and additionally certifies a common nested region. No unexplained constant is substituted into the approximation theorem to turn a finite-model statement into a diffusion claim. The approximation results and continuous-time economy are preserved; the numerical evidence identifies exactly the decisions for which a further target-error bound is consequential.

9. PARTICIPATION, TERMINATION, AND PORTFOLIO PERMISSIONS

The operating obligation and the investment permissions are distinct economic primitives. An agent who can leave a costly activity need not value an adjustment technology as does an agent committed to continue. Likewise, a change in the permitted size of a position need not have the same effect as a change in preferences. We embed the original operating economy in a participation contract, characterize the price of relaxing its obligation, and identify the investment-permission margins that move the two indifference loci. The original noncancellable experiment remains a member of this family.

9.1. *An Operating Mandate with Participation and Limited Enforcement*

At date zero, an agent may accept an operating mandate or take the outside payoff $G(x_0)$. Acceptance fixes the managed initial state $x_0 = (2, 1.25)$, the rebalancing calendar, and the stochastic settlement technology of Section 8.1. A separate, additively separable utility numeraire supports a signing grant q and an enforceable termination charge F . Neither payment is credited to managed wealth X . Crediting the grant to that state would change consumption, transitions, and the experiment, and is not the contract considered here.

The mandate stipulates a minimum operating term mh , where $h = 1/8$ and $m \in \{1, \dots, 8\}$. At an interior rebalancing date $n \geq m$, the agent may surrender for $G(x) - F$. Before that date, operation is compulsory. At a resource or preference boundary the mandate is discharged without a charge: the boundary records a contractually verifiable inability to continue within the permissible state domain. This discharge provision differs from an elective surrender at an ordinary interior node. It explains why the same underlying liquidation payoff does not imply the same right to terminate. The case $m = 8$ is the original noncancellable mandate, independently of F .

The charge is supported by externally enforceable guarantee capacity $F \leq E \leq \bar{E}$, where $\bar{E}$ is the available capacity limit. Let $C(E) \geq 0$ be the agent's nondecreasing ex ante cost of maintaining that capacity, measured in the separate numeraire. It may represent a surety premium or a collateral carrying cost. It is independent of the within-mandate control conditional on the contracted capacity, and enters the participation payment explicitly. No costless, unbounded borrowing against managed wealth is assumed. A fully collateralized interpretation is also possible if the collateral account and its return are specified separately; its carrying cost cannot be silently omitted. The calculations below first report $C(E) = 0$ so that the effect of the operating technology is visible. A specified positive $C(E)$ adds to the required grant whenever participation binds.

These are assumptions of a theoretical procurement economy, not claims about a particular intermediary. They provide a contracting interpretation and a transparent

participation test. They do not establish that this mandate, its settlement lottery, or its enforcement technology is optimal among all contracts. In particular, a principal must be willing to pay for the service; declaring the operation compulsory does not itself satisfy participation.

Write $r \in \{\text{adj}, 0\}$ for the two adjustment regimes and $\sigma \in \{+, -\}$ for the initial position class. The regime 0 prohibits deliberate adjustment, not stochastic movements of preferences. Let $W_\sigma^r(\lambda, d, F, m)$ denote the operating value before the signing grant and the capacity cost. A deterministic Markov policy p specifies both operating actions and an elective surrender rule. Let τ be its first discharge, surrender, or terminal date. Write $\mathcal{E}_p$ for elective surrender before discharge or maturity, and define

$$\mathcal{A}_p = \mathbb{E}_p \left[\int_0^\tau e^{-\rho t} dt \right], \quad \mathcal{S}_p = \mathbb{E}_p [e^{-\rho \tau} \mathbf{1}_{\mathcal{E}_p}], \quad (77)$$

$$J_p(\lambda, d, F, m) = \mathcal{B}_p(\lambda, m) + d\mathcal{A}_p(\lambda, m) - F\mathcal{S}_p(\lambda, m). \quad (78)$$

Here $\mathcal{B}_p$ contains consumption utility, preference-state utility, adjustment cost, and liquidation without the charge. In the finite protocol, the integral in (77) uses the actual within-period exit annuity. The surrender exposure is discounted incidence, not an undiscounted exit probability. Fixing a policy fixes these features; reoptimizing a contract need not do so.

PROPOSITION 21—Participation, service procurement, and the surrender envelope: *For fixed (λ, m) and a fixed feasible action technology, $W_\sigma^r = \max_{p \in \mathcal{P}_\sigma^r(m)} J_p$ is convex in (d, F) , nondecreasing in d , nonincreasing and 1-Lipschitz in F . The smallest nonnegative signing grant satisfying participation is*

$$q_r^*(E) = \left[G(x_0) - \max_\sigma W_\sigma^r + C(E) \right]_+. \quad (79)$$

Suppose a principal obtains flow b per unit of discounted operating duration and pays the grant, while termination charges are remitted to a separate enforcement institution. For a specified optimal agent policy p_r^ with $\mathcal{A}_{p_r^*} > 0$, that contract is individually rational for both parties at the minimum grant if and only if*

$$b \geq \frac{q_r^*(E)}{\mathcal{A}_{p_r^*}}. \quad (80)$$

At a point with a unique exposed feature vector, the class-value derivatives are $W_d = \mathcal{A}_{p^}$ and $W_F = -\mathcal{S}_{p^*}$. At a tie, the directional derivative is the maximum of $v_d \mathcal{A}_p - v_F \mathcal{S}_p$ over optimizing policies. These statements concern the specified contract, not an optimization over all mandates.*

The proof is an affine-policy envelope argument, given in Supplement S.13. Equation (80) imposes a substantive participation restriction on the principal as well as on the agent. It compares compensation with delivered service rather than interpreting a low grant for an immediately abandoned activity as a procurement improvement. The displayed values are utility-numeraire quantities. They are not monetary willingness-to-pay estimates for an investor whose entire wealth follows CRRA utility.

Define $\Delta^r = W_+^r - W_-^r$ and $\mathcal{O} = \Delta^{\text{adj}} - \Delta^0$. Where the feature derivatives exist,

$$\Delta_d^r = \mathcal{A}_+^r - \mathcal{A}_-^r, \quad \Delta_F^r = -(\mathcal{S}_+^r - \mathcal{S}_-^r). \quad (81)$$

Thus increasing a surrender charge need not increase either the relative adjustment option or the positive-position advantage. The sign depends on which class uses surrender more. If all class-specific participation constraints bind, then

$$\mathcal{O} = (q_+^0 - q_+^{\text{adj}}) - (q_-^0 - q_-^{\text{adj}}), \quad (82)$$

for a common capacity cost and outside payoff. The relative option is the difference between the compensation savings from adjustment in the two position classes. Adjustment weakly lowers each class's required compensation; it need not lower it more for the positive class.

9.2. When Does a Finite Charge Implement Continued Operation?

Let V_n^r be the original noncancellable continuation value. For the new mandate the recursion at eligible interior nodes is

$$W_n^r(x) = \max \left\{ G(x) - F, \max_{a \in A_n^r(x)} [r_n^{\lambda,d}(x, a) + K_n^\lambda(x, a) W_{n+1}^r] \right\}. \quad (83)$$

Before date m only the second entry is available. Boundary nodes continue to pay G , terminal nodes pay G , and the date-zero operating action retains the original positive-duration sign restriction. A charge for an early forced exit is not introduced.

Let R_n contain all live grid nodes reachable at date n from x_0 using any baseline action and either endpoint law, without voluntary surrender. This support set includes paths that no optimal policy uses. It is closed under feasible live transitions and works simultaneously for both regimes and for every $\lambda \in [0, 0.25]$.

PROPOSITION 22—A statewise enforcement frontier: *For a fixed contract and $m \geq 1$, define*

$$F_r^*(m; \lambda, d) = \max_{m \leq n < 8} \max_{x \in R_n \text{ interior}} [G(x) - V_n^r(x; \lambda, d)]_+, \quad (84)$$

with an empty maximum equal to zero. The voluntary-surrender and noncancellable values coincide at every reachable continuation node if and only if $F \geq F_r^$. This condition implies equality of both initial class values. It is sufficient, but need not be necessary, for equality only at the initial state.*

The threshold is nonincreasing in m and in d . Moreover,

$$F_{\text{adj}}^*(m; \lambda, d) \leq F_0^*(m; \lambda, d). \quad (85)$$

Consequently, deliberate preference adjustment weakly substitutes for enforcement capacity in implementing continued operation, even though its effect on the relative initial-position option can have either sign. If $\underline{V}_n \leq V_n^0$ uniformly on a parameter region, replacing V^0 by $\underline{V}$ in (84) gives a sufficient common charge for both regimes throughout that region.

The necessity in this proposition is statewise: an eligible node with $G - F > V_n$ makes equality at that node impossible. The sufficiency follows by backward induction. The order in (85) follows from inclusion of feasible policies, not from an assumed sign of the four-class option. This distinction matters in the numerical results. A low charge can cause the unadjusted positive class to use surrender disproportionately, reversing

the relative option even though adjustment still reduces the compensation required within every class.

The implementation of the regional frontier uses two feasible no-adjustment policies and their all-date Bernstein values. Nonnegative duration makes the lower benefit endpoint $d = 0.4$ sufficient. On each of sixteen law subintervals, the minimum restricted coefficient is a policy lower value at each node. Taking the larger of the two policy lower values is still a lower value for V^0 . Subtracting the arithmetic allowance and maximizing the resulting liquidation deficit over R_n supplies the common charge. This calculation does not require an optimizer at every law probability or a neural policy. It also avoids paying for enforcement at grid nodes that the initial mandate cannot reach.

Free surrender provides the opposite endpoint of this frontier. If G is a supersolution of every feasible one-step operating backup, backward induction gives $W_n = G$ at all surrender-eligible dates when $F = 0$. With the first interval compulsory, the agent then purchases only that interval of operation. With surrender additionally available at date zero, the original risk classification is bypassed and the value is simply $G(x_0)$. Neither counterfactual is an implementation of the noncancellable contract.

9.3. Investment Permissions and Their Shadow Values

The mandate can separately restrict funded purchases and stock borrowing at the initial date. A long permission L limits risky holdings to LX_0 ; a short permission S limits borrowed risky holdings to SX_0 . A liquidity-reserve requirement can motivate the former, and a stock-loan or collateral line the latter. Their levels need not be symmetric. The baseline permits $L = 0.8$ and $S = 0.5$. These numbers are contractual parameters, not estimates of market-wide restrictions. Both $\pi = L$ and $\pi = -S$ carry financial risk.

We vary these permissions only at the first date, leaving all later menus, the 1/8-year decision horizon, and the transition protocol unchanged. To distinguish a permission effect from a missed action-grid point, we allow every risky share in the relevant interval for each common consumption–adjustment pair. The three original first-date proposals are retained when feasible. Consumption and deliberate adjustment themselves remain on their original finite pairs; the later target is still the complete 1,568-action menu. This is a precisely defined enlargement of initial financial permissions, not continuous optimization of all three controls over the full horizon.

PROPOSITION 23—Exact first-date risky-share reduction: *Fix the optimized continuation V_1^* , a common pair (c, θ) , and initial state $x_0 = (2, 1.25)$. For $\pi \in [-0.6, 1]$, every initial proposed increment remains strictly inside the state domain. On this range, the first-date payoff $Q'(c, \theta, \pi)$ under the finite settlement protocol is continuous and piecewise affine in π . Its breakpoints are the risky shares at which a conditional arrival wealth coordinate crosses a wealth-grid node. Hence the nonpositive maximum over $[-S, 0]$ and the positive supremum over $(0, L]$ are obtained by checking endpoints and these breakpoints for each common (c, θ) pair, together with feasible frozen proposals. The closure at zero is used only for the positive supremum; a reported positive maximizer must be strictly positive.*

This reduction replaces a dense risky-share scan by a finite exact search for the specified enlarged initial menu. The numerical implementation contains 121 common consumption–adjustment pairs and 2,689 candidate triples over the wider range. It does not convert finite consumption and adjustment choices into continuous ones. Nor does it convert the piecewise-affine settlement payoff into the diffusion’s local mean–variance formula.

On a smooth interpolation cell at a binding optimum, let

$$\mu_L^r = \partial_\pi Q^r(c_+^*, \theta_+^*, L), \quad \mu_S^r = -\partial_\pi Q^r(c_-^*, \theta_-^*, -S). \quad (86)$$

These are one-sided values of relaxing the long and short permissions. At a switch between optimizing actions the appropriate directional envelope replaces a single derivative. For initial-only nested permission sets, the positive class value is nondecreasing in L , and the nonpositive class value is nondecreasing in S , regardless of whether a differentiable multiplier exists.

At a regular indifference locus with $D_A^r = \mathcal{A}_+^r - \mathcal{A}_-^r \neq 0$, the envelope identities give

$$\frac{\partial d_r^*}{\partial F} = \frac{S_+^r - S_-^r}{D_A^r}, \quad \frac{\partial d_r^*}{\partial L} = -\frac{\mu_L^r}{D_A^r}, \quad \frac{\partial d_r^*}{\partial S} = \frac{\mu_S^r}{D_A^r}. \quad (87)$$

These are local statements under the exposed-feature and nonzero-slope conditions, not global monotonicity assertions about a difference of convex functions in d . They separate three economic margins: duration values an operating benefit, surrender exposure values a termination charge, and the first-date continuation gradient values a binding portfolio permission. Preference adjustment changes these objects through reoptimization. It is not a substitute for observing which permission actually binds.

9.4. Certification of the Expanded Contract Family

Adding surrender is compatible with the transition-law certificate. It is an additional action with reward $G - F$ and zero live kernel. It therefore preserves affine dependence on the law probability, nonnegative substochastic kernels, and the finite-horizon information ordering. A fixed policy has the representation

$$J_p(\lambda, d, F) = \sum_{j=0}^8 b_{j,8}(\lambda) (B_{p,j} + dA_{p,j} - FS_{p,j}), \quad (88)$$

where $b_{j,8}$ denotes the degree-eight Bernstein basis. Early surrender merely shortens the effective polynomial degree; degree elevation retains the common representation. The upper chord and count recursions use the same expanded menu, including the same date and state eligibility of surrender. Feasible lower policies retain their own surrender choices when evaluated at another fee.

For each law interval, upper values at the four (d, F) corners give a tensor upper bound by convexity of the optimized value in these two reward parameters. Each fixed-policy lower value is affine in them. Restricted Bernstein coefficients then bound the two initial class differences uniformly on a three-parameter box. The maximum over lower policies is taken only after a bound valid for each individual policy has been constructed. The full argument, including arithmetic, is given in Supplement S.13.

There are three separate targets. The mathematical settlement contract uses real arithmetic; the implemented certificate concerns its stored transition and reward arrays; and the diffusion or another settlement contract is a different economic target. Let $\epsilon_{\sigma,n}^r$ be a bound for transferring a class value between two specified finite-array targets. A sufficient backward error account is

$$\epsilon_{\sigma,8}^r = \epsilon_G, \quad \epsilon_{\sigma,n}^r \leq \epsilon_{r,n} + \epsilon_{K,n} B_{n+1} + \beta_n \epsilon_{\sigma,n+1}^r, \quad (89)$$

where reward errors include surrender and boundary payments, ϵ_K is a row- ℓ^1 kernel error, B bounds continuation magnitude, and β bounds row mass. The two class errors must be added to the error in a difference certificate. The reported arithmetic allowance covers evaluation of the stored arrays, not an uncomputed ϵ_K for the exact constructor or for a diffusion approximation. Selected-control reconstruction checks consistency, but does not estimate an interval enclosure of every transcendental operation. No target transfer is inferred from a small reconstruction discrepancy.

10. CORRESPONDENCE WITH REGIONAL PARAMETER METHODS

The appropriate comparison is not between a reward-control method and literature confined to transition probabilities. [Quatmann, Dehnert, Jansen, Junges, and Katoen \(2016\)](#) define parametric Markov decision processes and reward properties. [Heck, Quatmann, Spel, Katoen, and Junges \(2025\)](#) develop dependency-sensitive parameter lifting for parametric Markov chains. The latter model class does not include the agent's action choice in the same way, but its treatment of shared parameters is directly relevant. We state the correspondence at the level of the target and information structure.

Write the finite Bellman operator as

$$(\mathcal{T}_n^\lambda v)(x) = \max_a \{r_n^\lambda(x, a) + K_n^\lambda(x, a)v\}, \quad (r_n^\lambda, K_n^\lambda) = (1 - \lambda)(r_n^0, K_n^0) + \lambda(r_n^1, K_n^1). \quad (90)$$

Augment a state by its date and add a cemetery state carrying the missing mass of a substochastic row. The absorbed state has zero subsequent reward; exit rewards already included in r_n^λ must not be paid twice. Set terminal rewards at date H . An action-dependent reward can equivalently be assigned to an intermediate action state. These transformations produce a finite-horizon parametric reward MDP. Signed rewards can be accommodated directly in finite-horizon Bellman comparisons; when a nonnegative-reward encoding is desired, a date-specific common shift must also be paid on absorbed paths so that its total is policy independent. These are model correspondences, not a priority claim or an assertion that two implementations are identical.

The distinctions are as follows. In the economic target, λ is common across dates and states, and the agent chooses the action before the endpoint-law draw. A rectangular relaxation permits endpoint choices to reset across state-date visits. It therefore loses a dependency that a fixed- λ policy must obey. The count-information recursion retains the total number of one endpoint's draws remaining, but does not reveal the next draw before the current action. Its maximization is over one common action for the conditional mixture, not separate actions chosen after observing which endpoint will occur. The extra count information yields an upper value for (90). Generalized lifting and dependency-reducing transformations address related losses of parameter dependence; their pMC results should not be silently identified with this controlled reward recursion.

Our corrected chord is a further compression of the count bound. Its signed cross-operator term is

$$(K_n^a - K_n^b)(V_{n+1}^b - V_{n+1}^a), \quad (91)$$

followed by positive backward propagation of a statewise correction. It replaces an interior count triangle by one correction vector per nontrivial date, while preserving an upper bound and the coefficient ordering established in Supplement S.10. On the lower side, actual feasible class policies are evaluated under the common parameter.

Their Bernstein coefficients preserve the dependency rather than receiving the upper relaxation’s extra information. This is why an upper relaxation and a neural training residual cannot substitute for feasible-policy evaluation.

The contribution claimed here is this particular signed compression, its ordering and total policy-bank error analysis, and its measured use in a common-target economic decision certificate. The comparison does not claim that regional expected-reward control, dependency-sensitive abstraction, or policy reuse first becomes possible with NBO. The count and chord timing comparison below is also not a benchmark against the complete software systems of [Quatmann, Dehnert, Jansen, Junges, and Katoen \(2016\)](#) or [Heck, Quatmann, Spel, Katoen, and Junges \(2025\)](#). It is an attributable comparison of the two mathematically matched upper components with the same economic target and lower evidence.

11. COMPUTATION AND INDEPENDENT POLICY EVALUATION

11.1. *The Stochastic Transition and Its Boundary*

For the portfolio economy we use four equiprobable weak-Euler branches. Independent signs $z_1, z_2 \in \{-1, 1\}$ generate

$$\Delta u = \theta h + \sigma_u \sqrt{h} z_1, \quad \Delta X = [(r + \pi(\mu - r))X - c]h + \pi X \sigma_X \sqrt{h} \{q z_1 + \sqrt{1 - q^2} z_2\}. \quad (92)$$

Thus both variances and the cross covariance are present. On each branch we follow the line segment from s to $s + \Delta s$ until its first boundary intersection, at fraction $\alpha \in [0, 1]$ of the step. The branch contribution is

$$r_k(s, a) \frac{1 - e^{-\rho \alpha h}}{\rho} + e^{-\rho \alpha h} \{1_{\{\alpha=1\}} v_{n+1}(s + \Delta s) + 1_{\{\alpha<1\}} G(s + \alpha \Delta s)\}. \quad (93)$$

An initial boundary state stops immediately and receives G . An interior step that hits the boundary exactly may equivalently use its boundary datum. No branch is returned to an interior state after an exit.

This polygonal stopped random walk is an approximation to Brownian stopping, not an exact Brownian first-passage simulation. The reference operator uses positive bilinear interpolation for live continuations. The neural training operator evaluates its frozen continuation network at those same physical successor locations. Policy evaluation subsequently uses a separate NumPy implementation with positive interpolation. These two approximations are compared rather than assumed to be identical.

Generator tests use $1, u, X, u^2, X^2, uX$. At an interior test point, their discrepancies have the expected drift-squared Euler remainder and recover the complete covariance matrix. A separate test verifies that opposite preference controls produce different conditional means even when the displacement is smaller than a grid cell. Interpolation preserves that difference; nearest-node rounding can erase it. The differential actor graph, finite-step actor graph, and immediate stopping boundary are also tested independently.

Table I reports a refinement sequence with $\max(\Delta u, \Delta X)^2/h$ decreasing from 0.04 to 0.01. This matters because interpolation error divided by the time step must be controlled in a generator approximation. The action mesh is held fixed in this sequence, and action refinement is assessed separately. These calculations diagnose discretization sensitivity; three meshes do not themselves constitute an error certificate for the continuous-time value.

TABLE I
JOINT STATE-TIME REFINEMENT OF THE STOPPED PREFERENCE MODEL

Grid	Dates	V_0	C_0	Seconds
17×25	4	-1.140548	0.009344	0.31
33×49	8	-1.136916	0.010335	2.07
65×97	16	-1.135019	0.010902	15.45

Initial state $(2, 1.25)$; $k = 2$. All rows optimize $5 \times 5 \times 7$ consumption, adjustment, and portfolio actions. C_0 is discounted integrated $\theta^2/2$. These are finite-approximation values, not certified continuous-time errors.

TABLE II
RE-SOLVED PREFERENCE-ADJUSTMENT COST PANEL

k	V_0	C_0	c_0	θ_0	π_0
0.5	-1.117524	0.014614	0.800	0.200	-0.500
2.0	-1.135835	0.009888	0.800	0.200	-0.500
8.0	-1.160291	0.001432	0.800	0.100	-0.500

Each row re-solves all dates and all three control coordinates on the same 25×37 grid, eight dates, and $9 \times 9 \times 13$ action mesh. The initial state is $(2, 1.25)$. The comparison concerns cumulative effort, not a universal pointwise policy ordering.

11.2. Preference-Cost Comparisons

The cost panel solves the entire finite problem again at $k = 0.5, 2, 8$, using a 25×37 state grid, eight dates, and $9 \times 9 \times 13$ controls. Consumption, preference adjustment, and portfolio share are all optimized. Discounted effort is evaluated under the resulting policy, including its effect on the exit time. Table II reports the initial state $(u, X) = (2, 1.25)$.

Discounted cumulative effort falls as k increases, consistently with Theorem 5. The portfolio and consumption coordinates at this initial state are at their bounds for all three displayed coefficients. This is not evidence for the interior formulas in (29). In particular, a short portfolio position in this calibration should not be described as an unconditional preference-hedging effect. The liquidation contract, the level of the running utility, and the portfolio's effect on exit jointly determine the optimum. Occupation-weighted exit probabilities are computed by forward propagation of the interpolation-defined Markov chain; an unweighted fraction of exiting grid branches would measure a different object.

11.3. Neural Policies in the Preference Economy

The ordinary neural proposal has two tanh hidden layers of width 32 and three bounded outputs. The safeguarded critic has two tanh hidden layers of width 64. Its output is $G(s) + (1 - t)m(s)N_{\xi}(s)$, where m vanishes on every face of the rectangle. The implementation uses separate L-BFGS fits, 2,048 uniformly sampled training states per seed, eight dates, and a feasible $5 \times 5 \times 7$ comparison set containing the actor's proposal. No grid reference values or optimal reference policies enter these training targets.

After training, the entire feedback sequence is frozen and evaluated on a 33×49 grid. The comparison reference uses the same number of dates and a finer $9 \times 9 \times 13$ action

TABLE III
NEURAL POLICIES AND INDEPENDENT EVALUATION IN THE STOPPED ECONOMY

Run	$V_R - J_0$	Critic RMSE	Critic max.	Gain max.	Seconds
Pilot 101	0.026720	0.066694	0.676974	1.340226	13.5
Safe 101	0.001018	0.047925	0.298592	0.228318	88.3
Safe 202	0.001366	0.044197	0.346920	0.210235	83.0
Safe 303	0.001177	0.052661	0.934824	0.228568	81.5

V_R is the separately solved 33×49 , eight-date, $9 \times 9 \times 13$ reference at $(2, 1.25)$. Critic errors compare the initial critic with its independently evaluated policy over the whole grid. Gain is the largest feasible one-step improvement over all interior states and dates against the evaluated continuation. The pilot has a smaller critic and training budget; it is not a matched ablation. Hard boundary discrepancies are evaluated separately in the raw record.

TABLE IV
FULL-VECTOR POLICY DISCREPANCIES AGAINST THE MATCHED REFERENCE

Run	MAE c	MAE θ	MAE π	Max. c	Max. θ	Max. π
Pilot 101	0.0136	0.0678	0.3212	0.562	0.400	1.300
Safe 101	0.0047	0.0165	0.1065	0.562	0.400	1.300
Safe 202	0.0047	0.0157	0.0954	0.562	0.400	1.300
Safe 303	0.0048	0.0158	0.1085	0.562	0.400	1.300

Mean absolute and maximum discrepancies include every interior common node and all eight dates. Coordinates have different economic units and are not added together or called a Hamiltonian gain. Both policies use their stated finite approximations; these are not errors against a continuous-time oracle.

mesh. A separate $7 \times 7 \times 11$ deviation search uses the *evaluated policy continuation*, not the learned critic, and includes the incumbent action. Consequently, a reported gain is a feasible payoff improvement. It is not a norm of the difference between two actions and is not an upper bound over the continuum of actions.

Table III reports all three safeguarded seeds and retains a lower-budget, unsafeguarded pilot. The pilot differs in width and optimization budget, so its comparison is a development diagnostic, not an isolated causal ablation. Table IV reports errors for the full control vector. A small error at the initial state does not imply a small maximum error near the boundary, and a policy coordinate can change substantially near a switch. The critic, policy value, and feasible-deviation columns therefore remain distinct.

The hard boundary architecture attains the displayed boundary condition to numerical precision. It does not certify the interior value or the Brownian stopping approximation. The sizable worst-case diagnostics identify where additional computation is needed for a uniform guarantee; they are not suppressed by averaging across states or seeds. Conversely, the initial-state comparison and the re-solved cost panel provide economic information that is not captured by the largest training residual alone.

11.4. Repair and Contract Counterfactuals

The policy errors above can be repaired without treating the learned critic as a value certificate. We freeze the three neural policy arrays and form, at each date and state, the union of the $9 \times 9 \times 13$ reference mesh, the $7 \times 7 \times 11$ diagnostic mesh, and the three frozen proposals. The common mesh contains 1565 distinct actions; the complete menu

TABLE V
NEURAL-POLICY LOSS BEFORE EVALUATED-CONTINUATION REPAIR

Seed	Feasible gain	Initial loss	Population loss	All-node loss
101	0.228318	0.001058	0.000730	0.236122
202	0.215071	0.001406	0.000783	0.245815
303	0.228568	0.001217	0.000800	0.236109

All losses are value differences in the same finite economy, relative to exhaustive backward optimization. The action menu is the union of the reference and original diagnostic meshes and three frozen neural proposals at each node. The initial state is $(2, 1.25)$; the population is uniform on grid nodes in $[1.6, 2.4] \times [0.8, 1.6]$. “All-node includes every date and state, not only occupied states. The repaired policy has zero remaining gain in the computed backups over this menu. The published diagnostic mesh is contained in the new menu.

has 1568 entries per node. Including the original diagnostic mesh is important: two different action meshes need not be nested merely because one has more points. Every previously tested profitable deviation is feasible in the repair calculation.

The target is now an explicit eight-date, 33×49 killed Markov chain. A branch’s live interpolation weights are nonnegative; its stopped payoff, discounted duration, and adjustment effort are immediate rewards. Backward maximization uses the *evaluated* next-date optimal continuation, and a separate fixed-policy recursion evaluates (B, A, C) . In this finite model, the incumbent proposals are feasible and the repaired policy weakly improves their values at every node. Table V gives the pre-repair errors. The post-repair Bellman gain over the entire stated menu, using independently evaluated policy continuations, is 4.441×10^{-16} , below 10^{-10} .

The largest initial-fit gains occur near the low-risk-aversion and high-wealth portion of the rectangle. Their exact dates, states, and occupation masses are saved. Under both the focal state and the stated initial population, the single worst-gain nodes have zero occupation mass. This explains why a large all-state error can coexist with a much smaller initial welfare loss; it does not make the global error disappear. The population losses and the occupation-weighted gains are distinct calculations, and the latter are not used as upper bounds on optimal-policy loss. The repair addresses all states rather than discarding the low-occupation ones.

At $k = 2$, the adaptive construction uses 47 anchors to cover every $d \in [0, 1]$. Its maximum two-policy chord gap is 0.00093070, below 10^{-3} at every date and state. The library evaluation does not retrain an actor or solve a new control maximization at a counterfactual query. A direct comparison at 101 parameter values finds a maximum value loss of 2.409×10^{-4} ; this comparison validates the implementation but is not the source of the continuum claim. The latter follows from the complete breakpoint enumeration in Section 5.

Table VI reports five independently optimized initial-state counterfactuals. Applying Proposition 8 to the full menu certifies a negative optimal initial share on $[0, 0.361017]$ and a positive share on $[0.373654, 1]$, with inward-rounded displayed endpoints. The intervening band is unresolved by this certificate. The sign conclusion uses action-value separation and therefore does not confuse a 10^{-3} global welfare tolerance with a 10^{-3} tolerance on an action coordinate.

The contract mechanism is economically specific. Raising the operating benefit rewards longer discounted operation and changes the portfolio from a short position toward a long one in the displayed economy. Adding the corresponding annuity

TABLE VI
OPERATING-BENEFIT COUNTERFACTUALS

d	Value	Duration	c_0	θ_0	π_0	Sign value gap
0.00	-1.136507	0.887826	0.80	0.20	-0.50	-0.000876
0.25	-0.914548	0.887930	0.80	0.20	-0.50	-0.000277
0.50	-0.692266	0.890233	0.80	0.20	0.80	0.000299
0.75	-0.469703	0.890536	0.80	0.20	0.80	0.000869
1.00	-0.246682	0.894175	0.80	0.20	0.80	0.001281

Initial state $(u, X) = (2, 1.25)$, $k = 2$. Duration is A^p , not the probability of survival. The final column is the best value among nonnegative-share first actions minus the best value among nonpositive-share first actions, using the optimal finite-model continuation. A positive entry favors the positive-share group. These five direct solves are distinct from the continuum certificate, which uses interval envelopes and fixed feasible action–continuation pairs.

amount to liquidation leaves the full policy correspondence unchanged; an independently evaluated terminal-exposure recursion verifies the accounting identity. The endpoint reversal also occurs when the shock correlation is set to zero or to 0.25. It is therefore not an unconditional sign prediction from negative preference correlation. The model does not claim that a unit operating benefit is empirically estimated or that this portfolio response holds under every liquidation contract.

Supplement S.8 reports joint state, time, and action refinements. The two endpoint signs persist, while the portfolio at $d = 0.5$ changes from negative on the coarsest economy to positive on the two finer ones. Value levels also move with resolution. This makes the location of a diffusion-model switch a separate approximation question; it does not invalidate the all-parameter statement for the specified finite target. The original continuous-time operator and its error-discrepancy term remain in the analysis.

For an actual 101-query workload, full direct solutions take 184.86 seconds, compared with 1.43 seconds for library selection and complete policy evaluation. Building the library and common kernel takes 87.47 seconds, including the interval-certificate calculation. These are incremental times given the same frozen proposals to both methods. The experiment measures a reusable family of complete policies and moments, rather than extrapolating from a single network forward pass. The stored library contains finite-state policy values; its construction does not establish a neural compression advantage in this low-dimensional economy.

11.5. Coupled Stochastic Resource Allocation

To examine a genuinely dynamic shared constraint, let $x_n \in \mathbb{R}^d$ denote productive stocks and $a_n \in \mathbb{R}_+^d$ investment. The three-period model is

$$x_{n+1} = A_d x_n + a_n + e_{n+1}, \quad A_d = 0.8I + 0.04(I_{+1} + I_{-1}), \quad \sum_i a_{ni} \leq 0.25d. \quad (94)$$

Here I_{+1}, I_{-1} denote the adjacent off-diagonal matrices. The shock has four equally likely realizations, $e = z_1 \ell_1 + z_2 \ell_2$, where $\ell_{1i} = 0.06$ and ℓ_{2i} is evenly spaced from -0.04 to 0.04 . Its covariance is rank two and generally has nonzero cross-stock entries. Heterogeneous initial stocks, neighbor spillovers, a common budget, and the aggregate cost couple the state coordinates.

TABLE VII
COUPLED STOCHASTIC RESOURCE ALLOCATION AT A COMMON ACCURACY TARGET

d	NBO upper loss	Proposal only	NBO sec.	Reference sec.	Passed
4	0.000645	0.01190	1.737	0.172	3/3
8	0.000461	0.09541	2.241	0.190	3/3
16	0.000567	0.06807	3.180	0.226	3/3

Loss columns are worst held-out cost-loss upper bounds across three seeds and 32 initial states per seed. Proposal only removes deployment improvement from the same fitted actor weights. Timing is the median across seeds at a 10^{-3} cost-loss target; NBO includes fitting and full evaluation, and the reference solves the same initial-state batch to its 10^{-3} gap target. A separate 10^{-7} reference gap is used for auditing, not for the timing comparison. The reference is a convex nonanticipative scenario tree, not a sparse grid.

The objective is expected total cost,

$$C^\pi(x) = \mathbb{E}_x^\pi \left[\sum_{n=0}^2 \left\{ q(x_n) + \frac{0.1}{d} \|a_n\|^2 \right\} + q(x_3) \right], \quad (95)$$

$$q(x) = \frac{\|x - \mathbf{1}\|^2}{d} + 0.2 \left(\frac{\mathbf{1}^\top (x - \mathbf{1})}{d} \right)^2.$$

This is a finite stochastic economy, not a discretization claim about a different diffusion. All 64 terminal shock paths are used when a policy is evaluated. The independent reference optimizes a nonanticipative scenario tree with 21 decision nodes per initial state. Its objective is convex quadratic and its nodewise resource constraints are genuine restrictions on investment.

The cost critic is

$$\hat{C}_n(x) = q(x) + b_n + \ell_n^\top (x - \mathbf{1}) + \sum_{j=1}^J w_{nj} [h_j^\top (x - \mathbf{1}) + b_j]_+^2, \quad w_{nj} \geq 0. \quad (96)$$

The fixed hidden features include coordinate, aggregate, and random directions; $J = 6d + 102$. Output coefficients are fitted by constrained least squares to the separated Bellman targets. A width-32 tanh proposal network uses a slack-softmax feasibility map and is trained against the improved controls. At deployment, projected convex improvement uses the analytic critic gradient, checked against automatic differentiation, and the gap (15). Squared-ReLU features are suitable here because the finite-step operator needs a value and an action gradient, not a classical second-derivative residual.

For each of $d = 4, 8, 16$ and seeds 101, 202, and 303, training uses 1,024 draws from $[-0.25, 1.65]^d$. Evaluation uses 32 fresh initial states from $[0, 1.2]^d$ and their complete shock trees. The reference is solved both to the common cost tolerance 10^{-3} for timing and to a gap below 10^{-7} for independent validation. Formula (16) gives a numerical upper bound on each tested policy's cost loss. These are double-precision convexity certificates; recorded feasibility and gradient discrepancies accompany them, rather than an assertion of interval-arithmetic proof.

Every displayed run meets the common cost-error target at its tested initial states. The shared constraint binds frequently, including at most or all initial states depending on dimension and seed. Thus the experiment tests a dynamically consequential budget rather than an inactive constraint. The same trained proposal networks are

TABLE VIII
REPRESENTATION AND ACTOR-FREE COMPARISONS

d	Actor-start loss	Zero-start loss	Quadratic loss range	Actor-zero difference
4	6.446×10^{-4}	6.446×10^{-4}	2.065×10^{-3} – 3.203×10^{-3}	1.256×10^{-10}
8	4.607×10^{-4}	4.607×10^{-4}	1.668×10^{-3} – 6.692×10^{-3}	1.104×10^{-10}
16	5.665×10^{-4}	5.665×10^{-4}	5.444×10^{-3} – 6.777×10^{-3}	2.679×10^{-11}

Each dimension includes seeds 101, 202, and 303 and the same 32 initial states per seed. The first two columns give the largest certified lifetime cost-loss upper bound across those seeds. The quadratic column spans the three maximum losses of a separately trained full positive-semidefinite quadratic critic. All use the same feasible controls and full shock trees; a reference Frank–Wolfe gap below 10^{-7} is added to each measured cost difference. Actor and zero starts use identical frozen nonlinear critic weights.

also evaluated without the deployment safeguard. This matched ablation, reported in the supplement, isolates the effect of the improvement calculation without changing network weights or training data.

At this query batch and tolerance, the specialized convex reference is faster end to end than training and evaluating the neural representation. The latter delivers a reusable feedback approximation, but a crossover in amortized cost requires a separate measurement. The experiment establishes accurate constrained neural control for this model and a transparent cost–accuracy comparison; it does not establish superiority over a solver that exploits the model’s convex structure.

11.6. *What the Actor and the Continuation Representation Contribute*

The actor does not enter the resource experiment’s critic targets: the target controls are computed before actor distillation. A same-weight comparison must therefore separate the value of continuation approximation from the value of actor initialization. We reload all nine checkpoints and solve the same convex improvements from either the actor output or zero. The largest complete-policy cost difference between these two arms is 1.256×10^{-10} . Both attain the stated 10^{-3} cost tolerance on every original held-out state. Actor initialization can save iterations, but those savings do not by themselves imply that its training cost is amortized.

The nonneural comparator is a full convex quadratic, $q(x) + b + \ell^\top(x - \mathbf{1}) + (x - \mathbf{1})^\top H(x - \mathbf{1})$ with $H \succeq 0$. It is fitted by projected least squares on the same training-state distribution and used in its own backward recursion. Unlike a diagonal quadratic, it permits every cross-state term. Its projected-gradient stopping diagnostic and Hessian eigenvalues are recorded. Table VIII shows that the richer squared-ReLU continuation class attains the common welfare target whereas these quadratic fits do not. This is evidence for the representational flexibility used here, not a theorem that a neural representation dominates all nonneural approximation classes.

For comparison with the previous execution, we retain the fresh sequence of initial states at query volumes 32, 128, 512, and 2,048. No initial state is added to the training set. Each query includes the complete endogenous policy tree and its 64 terminal shock paths. Table IX charges for critic training, zero-start online improvement, and policy evaluation, and compares that sum with the specialized scenario-tree solver at the same 10^{-3} cost tolerance. The independent tighter reference remains solely an accuracy check for both methods. At the largest workload the measured end-to-end cost ratio is 1.67, with a maximum policy-loss bound of 6.754×10^{-4} . The intermediate workloads

TABLE IX
FRESH-STATE QUERY WORKLOADS AT A COMMON COST TOLERANCE

Queries	Max. loss bound	Training	Query	Reference	Cost ratio
32	6.024×10^{-4}	0.812	0.040	0.152	0.18
128	6.574×10^{-4}	0.812	0.119	0.420	0.45
512	6.574×10^{-4}	0.812	0.600	1.477	1.05
2048	6.754×10^{-4}	0.812	2.600	5.708	1.67

Dimension four, seed 101; held-out state seed 541021, uniform on $[0, 1.2]^4$. Larger workloads extend the same fresh-state sequence. Times are seconds. Query time includes zero-start convex improvement and all 64 terminal paths per initial state. Reference time is for independent nonanticipative optimization to the common 10^{-3} cost tolerance. The cost ratio is reference time divided by critic training plus query time. Each reported maximum policy-loss bound is below 10^{-3} . Timings are medians of three evaluations; the separate tighter validation reference is not charged to either competing solver.

document that historical crossover against the original reference. They do not give the reference the reusable quadratic representation used below.

The historical crossover does not identify nonlinear continuation as the preferred reusable object. All these queries share a quadratic objective Hessian and an affine map from initial state to its linear term. Section 11.9 gives the conventional solver that same opportunity for reuse and changes the computational conclusion. The actor’s separate distillation time, initialization time, optimizer iterations, and projected break-even counts remain in Supplement S.8; the projections remain projections rather than new measurements. The original table is preserved as an executed comparison, not promoted into an advantage over the stronger reusable-QP baseline.

11.7. Smooth and High-Dimensional Controls

An additional stochastic LQR experiment uses correlated Gaussian shocks, neighbor coupling in the transition, and a rank-one cross-state cost at dimensions 4, 8, and 16. Its critic has quadratic features, its actor is linear, and its independent reference is Riccati recursion. This structured control checks backward evaluation, covariance contributions, actor coefficients, and full-policy values. It is not an experiment with a generic multilayer critic, nor a substitute for the constrained resource application.

Computational cost depends on the operator and representation. For a differential critic and rank- r loading B , the exact trace can be evaluated using r directional Hessian contractions. A K -probe trace instead uses K such contractions after forming the probe directions. Their total costs include network differentiation, loading multiplication, action optimization, sampling, and the iterations needed to achieve an error target. Per-iteration differentiation does not establish a polynomial solution-complexity theorem, and no threshold such as dimension 50 follows from that calculation. The semigroup experiments report their own costs without attributing them to unexecuted stochastic-Hessian acceleration.

11.8. Same-Oracle Optimistic Support and Neural-Action Attribution

Table X compares two coverage objectives with an identical exact Bellman oracle, action menu, state space, and tolerance. The initial-distribution version concentrates all weight on $(u, X) = (2, 1.25)$ at date zero. The statewise version takes the maximum gap

across all 1,617 states and all eight decision dates. Both start from $d = 0, 1$, insert a worst remaining corner, and use independently evaluated policy features. We report common kernel construction separately; bank memory counts policies and features, excluding shared transition arrays. The initial-distribution target needs only three anchors. Its

TABLE X
EXACT-ORACLE SCALAR OPTIMISTIC SUPPORT

Coverage objective	Anchors	Seconds	MiB	Global bound
Initial state/date	3	5.82463	1.14732	0.0529688
All states/dates	47	91.2708	17.9746	0.000930698

Notes: Same exact full-menu oracle and 10^{-3} target. The initial-state bound is 0.000410746. Global means all finite states, dates, and $d \in [0, 1]$. Common kernel setup costs 2.78081 seconds and is excluded from both preparation columns. MiB counts policies and features, not shared kernels or upper buffers. The independent full-line/adjacent-line discrepancy is 4.44089e-16.

much larger all-state/date bound demonstrates why that guarantee cannot be silently used for a different population or a later date. The statewise calculation uses 47 anchors. An independent calculation with every policy line reproduces its bound and verifies the adjacent-policy identity to floating-point tolerance. This is an executed comparison of guarantee scope and the scalar-OLS correspondence, not evidence that a new name reduces the oracle count.

We then remove every neural proposal from the deployed lower policies, leaving the same 1,565-action common mesh. The upper information continues to come from the original full, neural-inclusive target. Table XI reports the anchor losses and the maximum over every contract interval. The mesh-only bank remains within 10^{-3} of the *original* target for all $d \in [0, 1]$, states, and dates. This is not a comparison against an easier restricted optimum, and it is not an autonomous mesh-only anchor-discovery experiment: the 47 locations and the full-target upper oracle are shared. The neural

TABLE XI
REMOVING NEURAL ACTIONS FROM THE FEASIBLE POLICY BANK

Feasible lower policies	Anchors	Anchor loss	Interval bound
Full menu	47	0	0.000930698
Common mesh only	47	3.87125e-05	0.000930698

Notes: Both rows use the original neural-inclusive optimal upper target at the same anchors. The second row removes all three neural candidates only from the lower policies. The interval bound covers all states, dates, and $d \in [0, 1]$. Mesh-policy construction costs 88.0996 seconds; this is not an autonomous mesh-anchor-discovery time.

actions have a small, nonzero value contribution at some nodes, but are unnecessary for this tolerance. Exact dynamic programming constructs the repaired bank; independent policy evaluation supplies its features; the upper oracle supplies the welfare benchmark; policy switching supplies deployment. The initial trained proposals and their profitable deviations remain reported. The transition-law theorem is an additional guarantee applicable to either neural or mesh policies; it does not require reassigning these observed roles to learning.

11.9. A Reusable Structural Solver for the Resource Economy

Stack the 21 nonanticipative node controls into $z \in \mathbb{R}^{21d}$. For every initial state x , the same model has cost

$$C_x(z) = \frac{1}{2}z^\top H z + (F x + f_0)^\top z + c(x), \quad z_h \geq 0, \quad \mathbf{1}^\top z_h \leq 0.25d. \quad (97)$$

The matrices H, F and vector f_0 are assembled once from the affine state paths. We scale the controls by their node probabilities, compute the resulting Hessian’s largest eigenvalue once, and apply accelerated projected minimization. The scaling is constant within a node, so the node’s feasible-set projection is unchanged. Each query starts at zero controls. The computation uses no future-shock information, stored query answer, or cross-query warm start.

The comparison follows the structural counterexample in the fifth-round advisory report, with an explicit independent assembly in the revision source. We check the assembled costs and gradients against the original full-tree routine. Every query is evaluated on all 64 terminal paths, and its stopping gap is recomputed from the original tree gradient, not just the assembled matrix. Table XII uses the same fresh initial states and 10^{-3} target as the earlier workload. Setup is included for the QP; critic-only retraining, zero-start improvement, and full policy evaluation are included for the learned arm. Query timings are medians of three serial single-thread runs. The

TABLE XII
FRESH QUERIES WITH REUSABLE QUADRATIC STRUCTURE

Queries	Learned total	QP total	QP query	QP max gap
32	0.884792	0.00759194	0.00434819	0.000633261
128	0.964309	0.014003	0.0107592	0.000664162
512	1.57165	0.0405753	0.0373315	0.000664162
2048	3.66748	0.143248	0.140004	0.000677426

Notes: Times are seconds, single-thread, query medians of three. The learned total includes critic retraining and full policy evaluation; QP total includes structural setup and independent full-tree gradient/gap evaluation. All queries start at zero controls. The largest learned policy-loss upper bound is 0.000675399. The original reference timings and all timing samples remain in the JSON record.

reusable QP is faster in these workloads, including online time alone. The complete-tree gaps certify its costs at the same tolerance. This is a substantive change in the computational conclusion, not a revision of the historical elapsed times. A fitted PSD-quadratic *continuation approximation* and the exact parametric quadratic *program* are different comparators. The recorded failure of the former does not imply a failure of the latter. Consequently this resource model supports controlled neural approximation and component attribution, but supplies no empirical basis for choosing learning over the reusable structural solver. The additional transferable guarantee studied next concerns a nonlinear stopped economy with changed transition laws, not a retiming of this quadratic example.

11.10. A Continuum of Shock-Law Counterfactuals

Let law 0 be the original one-step quadrature with shock correlation -0.25 , and law 1 the otherwise identical quadrature with correlation 0.25 . At every date, law 1 occurs

with probability λ , independently across dates and without being known before that date's control. The state transition, exit payment, and discounted continuation are the corresponding mixtures. The state grid, consumption and portfolio bounds, adjustment menu, terminal payoff, and three frozen neural proposals remain unchanged. This is a change in the frequency of positive versus negative co-movement regimes in the finite economy. It is not the claim that interpolating two quadratures equals discretizing every intermediate Brownian correlation.

For each of the contract contrasts $d = 0, 0.5, 1$, we solve endpoint problems and adaptively subdivide the transition-parameter interval using Theorem 10. The endpoint policies are evaluated by (53). Eight closed Bernstein cells within each anchor interval bound its full policy-value gap; no target problem is solved at those cell endpoints merely to label a grid uniform. Table XIII reports the maximum certificate over every state, date, and $\lambda \in [0, 1]$. The exact parameter remains common across the horizon. The certificate is distinct from the older reward interval: the displayed d values are three fixed contracts, not a new joint continuum claim over (d, λ) . The count column

TABLE XIII
UNIFORM POLICY REUSE ACROSS TRANSITION MIXTURES

d	Anchors	Corrected bound	Count bound	Prep. seconds
0	19	0.000904573	0.0127756	52.5879
0.5	18	0.000899829	0.0123306	58.5677
1	17	0.000941412	0.00928855	53.1281

Notes: Each row covers every $\lambda \in [0, 1]$, all 1,617 states, and eight decision dates at the displayed fixed contract. Corrected preparation includes endpoint solves, policy-coefficient construction, and interval certification, excluding the common two-kernel build (5.43095 seconds) and subsequent independent replay. The count bound uses its separate informed upper oracle and three feasible anchor policies. It is valid but does not meet the 10^{-3} target. No joint (d, λ) -continuum or diffusion guarantee is asserted.

in this retained table uses one global interval and only the three feasible policies at $\lambda = 0, 0.5, 1$. The corrected column uses adaptive localization and 17–19 anchor policies. Both columns are valid, but their difference confounds the upper construction, localization, and lower-policy coverage; it is not evidence of a sharpness advantage for the correction. Section 11.12 supplies the matched comparison and the work account. Proposition 12 establishes that, on the same intervals and policy bank, localized count is weakly sharper.

Every final anchor interval is independently replayed at its endpoints and midpoint using a complete backward optimization of the same mixture model. Fixed-policy polynomial values are also checked against a direct selected-kernel recursion. Small independent models enumerate all regime sequences, and two-date models enumerate every deterministic Markov policy. These tests distinguish a correct polynomial identity from an accidentally favorable main-model result. The uncorrected-chord negative control has endpoint values zero and midpoint value $1/4$; some valid correction or alternative upper oracle is therefore necessary. This negative control does not establish the necessity of this particular correction among valid constructions.

All calculations use double precision. The transition arrays are positive, the prescribed finite actions are exhaustively checked, and recorded arithmetic discrepancies are compared with explicit tolerances. These are finite-model numerical certificates, not directed-rounding interval arithmetic or a diffusion discretization guarantee. A CSR

representation accelerates multiplication by the identical transition rows; its values are checked against the inherited array implementation, while fixed-policy evaluation retains a separate arithmetic path.

11.11. *The Preference-Formation Contribution to Risk Choice*

Table XIV evaluates the risk-class comparison in Section 7. Both arms retain preference shocks, the original covariance, the same operating boundary and settlement, and the full consumption and portfolio menus. The restricted arm sets deliberate adjustment to zero at *all* states and dates. The difference $\Omega_+ - \Omega_-$ is computed from separately optimized and independently evaluated risk-class policies, not from a first-order condition imposed on a binding control. At $d = 0.5$, the adjustment-enabled economy favors

TABLE XIV
DELIBERATE PREFERENCE FORMATION AND RISK-CLASS VALUES

d	Δ^{adj}	Δ^0	$\Omega_+ - \Omega_-$
0	-0.000875905	-0.00133454	0.000458639
0.25	-0.000276562	-0.000730716	0.000454155
0.35	-4.62573e-05	-0.000489849	0.000443592
0.365	-1.17117e-05	-0.000453738	0.000442026
0.375	1.13188e-05	-0.000429663	0.000440982
0.4	6.88949e-05	-0.000369477	0.000438372
0.5	0.000299199	-0.000135684	0.000434883
1	0.00128126	0.000950467	0.000330789

Notes: Positive Δ favors a positive first risky position; negative Δ favors the nonpositive class. Both classes optimize every subsequent action. The restricted arm sets deliberate adjustment to zero at all states and dates but retains preference shocks and every financial and stopping primitive. Each reported value is independently evaluated in the same cardinal utility units.

a positive first risky position, whereas the otherwise identical economy without deliberate adjustment favors the nonpositive class. The same disagreement occurs at $d = 0.375$ and $d = 0.4$. Thus the relative preference-formation option is large enough to reverse the risk ranking under an unchanged stopping contract. At $d = 1$, both economies favor the positive class: the operating-contract incentive can produce a reversal without deliberate preference formation, but the formation option changes where that reversal occurs. This separates a generic duration incentive from a preference-specific ranking effect through (65). Neither fixed consumption at a bound nor movement between endpoint portfolios alone supplies this identification.

Table XV holds the financial and stopping primitives fixed and varies the adjustment-cost coefficient. It reports opposite-sign brackets and the attaining policies' duration and effort differences. The ratio in (63) is a local regular-branch prediction in cardinal flow-utility units per unit of adjustment cost, not a global derivative through every policy kink. We do not claim uniqueness from a bisection bracket; uniqueness requires the uniform duration-separation condition of Proposition 14. The no-adjustment bracket separately shows which part of the risk response remains when the formation option is removed. Increasing the adjustment-cost coefficient moves the observed opposite-sign bracket toward higher duration compensation. The no-adjustment bracket lies further

TABLE XV
RISK INDIFFERENCE AND THE DURATION-EFFORT COMPARISON

k / restriction	Opposite-sign bracket	$A_+ - A_-$	$C_+ - C_-$	Ratio
0.5	[0.34687805, 0.34689331]	0.00230304	4.6285e-05	0.0200973
2	[0.37007141, 0.37008667]	0.00230304	2.12312e-05	0.00921875
8	[0.48182678, 0.48184204]	0.00230304	7.90303e-05	0.0343156
No adjustment	[0.55891418, 0.55892944]	0.00230304	0	0

Notes: Each bracket has opposite risk-advantage signs at its endpoints and width at most 2×10^{-5} . It proves existence of a crossing, not global uniqueness. Features and their ratio are evaluated at the midpoint. The ratio predicts a regular local branch slope only under the stated nonzero-duration and differentiability conditions. A zero-adjustment policy has zero adjustment effort. Bracket endpoints and their signed values are stored, rather than inferred from rounded table entries.

to the right. These results accord with the positive effort–duration ratios of the attaining class policies, subject to the local regularity conditions. The duration–effort and option-value propositions organize these optimized-policy comparisons. The primitive condition in Proposition 16 supplies a separate predictive result, and Proposition 17 supplies the class-specific transition-law certificate used below. Changing the operating boundary or settlement curvature would define further economic contracts, rather than innocuous numerical robustness checks. Their effects are not identified by the present matched restriction. The original boundary calculation, correlation panel, and joint numerical refinements remain available and are not silently replaced by the new experiment.

11.12. Matched Precision, Work, and Anchor Demand

We now hold fixed what differed in the retained global comparison. For every one of the 51 original anchor intervals, the corrected chord, localized count recursion, and rectangular upper recursion use the same two endpoint policies and the same eight closed Bernstein cells. Every anchor is freshly solved on the full 1,617-state, eight-date, 1,568-action target, and its policy coefficients are independently reevaluated. The shared values and coefficients reproduce their R6 records to the deposited numerical tolerance. The three full-bank rows therefore reproduce the sixth-round matched countercomparison while adding a common-machine work account. Two additional $d = 0.5$ banks retain every second or fourth anchor and include the right endpoint.

The localized count certificate is sharper in every full-bank row, as Proposition 12 requires. The compressed construction buys smaller upper-oracle work rather than smaller loss. At eight dates its 26 action-wide kernel applications per interval compare with 70 for the cached count recursion and 16 for the rectangular relaxation. The persistent correction needs seven additional state vectors, compared with 28 interior count vectors; neither number includes the shared endpoint values or the uncompressed lower-policy bank. The rectangular relaxation is cheaper in kernel work but loses much more precision here by allowing repeated resets of the regime probability. Thus the table separates a validity result, a sharpness result, and a cost result.

All elapsed-time columns include kernel construction, the required anchor solves, policy evaluation, local coefficient restriction, upper construction, and subdivision. Shared stages are measured once and attributed equally. The compact implementation expands one date’s upper coefficients during certification; this work is included rather

TABLE XVI
MATCHED UPPER-ORACLE PRECISION AND WORK

d	Anchors	Upper oracle	Bound	Upper sec.	Total sec.
0	19	Compressed	0.000904573	27.02	53.34
		Local count	0.000733846	95.12	121.40
		Rectangular	0.0187608	21.62	47.90
0.5	18	Compressed	0.000899829	25.46	51.01
		Local count	0.000685103	88.91	114.43
		Rectangular	0.0171499	20.01	45.52
0.5	6	Compressed	0.00698458	7.34	19.26
		Local count	0.00338132	25.61	37.52
		Rectangular	0.0427253	5.75	17.66
0.5	10	Compressed	0.00327336	13.35	29.79
		Local count	0.0020246	46.32	62.74
		Rectangular	0.0255638	10.35	26.77
1	17	Compressed	0.000941412	23.64	47.04
		Local count	0.000435745	81.90	105.27
		Rectangular	0.0150829	18.24	41.60

Notes: All states, dates, and $\lambda \in [0, 1]$; identical two-policy lower banks and eight cells within each compared interval. The 19-, 18-, and 17-anchor banks are the original R6 banks; the 6- and 10-anchor banks coarsen the $d = 0.5$ bank. Total time includes common kernel setup, anchor solves, policy evaluation, coefficient restriction, upper construction, and certification. Shared stages are attributed equally. One serial single-thread pass; diagnostic replay is excluded equally. Kernel-call and persistent-array counts, every interval bound, and stage times are in the replication records.

than removed from the timing. These are one-pass serial single-thread measurements, not medians of independently rebuilt machines. The operation counts provide a reproducible work comparison independent of elapsed-time noise. No new neural training or learned-speed advantage is inferred from them.

Common banks do not determine how many anchors an independently refined method would request. Table XVII therefore starts three procedures at $[0, 1]$ with empty per-arm caches, fixes $d = 0.5$, and uses the same left-first dyadic subdivision and 10^{-3} target. The chord-only and count-only arms use their own certificates to decide whether to split. The cascade tries count only after the chord fails. Every rejected parent interval, newly solved anchor, lower evaluation, and cell test is included in that arm's cost. This is an autonomous refinement experiment, unlike the coarsened-bank rows of Table XVI. The chord-only procedure requests 18 anchors; local count requests 11; the cascade

TABLE XVII
AUTONOMOUS REFINEMENT FROM EMPTY POLICY CACHES

Rule	Anchors	Final bound	Kernel calls	Total sec.
Chord	18	0.000899829	858	73.41
Count	11	0.000924471	1330	115.38
Cascade	11	0.000924471	1614	127.73

Notes: Full target, $d = 0.5$, tolerance 10^{-3} . Each arm starts at $[0, 1]$ and includes all rejected parent intervals. Kernel calls are action-wide upper-oracle applications, not anchor-solve calls. Total time includes both. Cascade tries local count only after compressed certification fails. All arms use identical dyadic split order, two endpoint policies, and eight certificate cells.

requests 11. Thus the cascade avoids 7 endpoint optimizations relative to chord-only refinement at the same certified tolerance. Its elapsed time is 127.73 seconds, against 73.41 and 115.38 for the two single-oracle procedures, respectively. The measured costs, rather than the anchor reduction alone, determine the computational comparison. The cascade's anchor set is checked to be a subset of the chord-only set, as the nesting argument predicts. It need not be the fastest procedure: buying a sharper certificate also consumes work. These finite-model results support a choice among upper-information budgets; they do not establish a dimension-free neural advantage or the optimality of any particular refinement strategy.

11.13. *A Certified Preference-Adjustment Effect on a Region*

The economic certificate holds the operating boundary, settlement function, consumption menu, financial technology, and stochastic preference shocks fixed. At $(u, X) = (2, 1.25)$ and $k = 2$, consider the whole rectangle

$$(\lambda, d) \in [0, 0.25] \times [0.4, 0.45].$$

The parameter λ is the same per-period probability of the positive-co-movement quadrature regime used above, not a new reward coefficient. Each risk class is solved at the four corners with and without deliberate adjustment. Their feasible policies are then evaluated as transition polynomials with affine contract coefficients. Localized count upper values at the contract endpoints are interpolated in d using convexity of the optimal class value. Proposition 17 certifies the resulting tensor coefficients on the entire rectangle.

TABLE XVIII
A JOINT-REGION CERTIFICATE FOR THE ADJUSTMENT EFFECT

Adjustment regime	Lower Δ	Upper Δ	Max. class gap
Adjustment available	6.88873×10^{-5}	0.000324537	7.5953×10^{-9}
No deliberate adjustment	-0.000369485	-5.29937×10^{-5}	2.19355×10^{-6}

Notes: Initial state $(u, X) = (2, 1.25)$, $k = 2$, full finite target, entire $(\lambda, d) \in [0, 0.25] \times [0.4, 0.45]$. Bounds include the arithmetic allowance. Maximum class gap is the larger of the two class-specific upper-minus-feasible-lower certificates in that regime. The per-class arithmetic allowance is 3.79765×10^{-9} ; the lower bound on the relative adjustment option is 0.000121881. Eight corner optimizations produce sixteen feasible class policies; eight count constructions certify the rectangle. No generic 10^{-3} welfare tolerance is used for these signs.

The positive first-risk class is strictly preferred throughout the region when adjustment is available, and the nonpositive class is strictly preferred throughout the same region when it is unavailable. Both signs survive the separate numerical allowance. Their difference bounds the relative adjustment option from below, so the conclusion identifies an incremental preference-formation effect under unchanged financial and settlement primitives. It is not the assertion that adjustment is necessary for every duration-induced risk reversal. The earlier $d = 1$ comparison, where both regimes select positive risk, remains in the paper.

This calculation uses the transition representation to certify the economically relevant decision. The upper-information and feasible-policy errors are measured separately

for the four class values; they are not assigned the generic 10^{-3} welfare tolerance. The tensor coefficients and a coefficient-only replay accompany the policy bank. Supplement S.11 gives the round-to-nearest arithmetic account relative to the stored finite model. Independent corner replay is an implementation check, not a replacement for the region inequality. The result concerns this initial state and rectangle, not all initial states, uniqueness of a global switching threshold, or a continuous-time discretization theorem.

12. ECONOMIC ROBUSTNESS AND THE COST OF A DECISION CERTIFICATE

The preceding experiments establish the operator and transfer comparisons on their original targets. We now hold the economic decision itself fixed when comparing the upper components, and then vary the economic primitives with both adjustment regimes reoptimized. Every historical comparison remains in the preceding tables. The additional evidence neither substitutes a generic welfare tolerance for a sign margin nor removes the original region when its interpretation is challenged by a perturbed transition technology.

12.1. *The Original Region and Its Moving Boundary*

TABLE XIX
THE ORIGINAL DECISION CORNER UNDER WEALTH-LOTTERY PERTURBATIONS

Wealth nodes	Menu	Δ^0	Δ^{adj}	Relative option
49	Common	-5.3001×10^{-5}	3.2453×10^{-4}	3.7753×10^{-4}
49	Full	-5.3001×10^{-5}	3.2453×10^{-4}	3.7753×10^{-4}
97	Common	6.1263×10^{-5}	4.2641×10^{-4}	3.6515×10^{-4}
97	Full	6.1263×10^{-5}	4.2641×10^{-4}	3.6515×10^{-4}
145	Common	9.7943×10^{-5}	4.6009×10^{-4}	3.6215×10^{-4}
145	Full	9.7943×10^{-5}	4.5954×10^{-4}	3.6160×10^{-4}

All rows retain 33 preference tiers, eight dates, and $(\lambda, d) = (0.25, 0.45)$. Common uses the identical 1,565-action union. Full also uses the three unchanged proposals at original nodes and their specified wealth interpolation at new nodes. Both classes and both adjustment regimes are reoptimized. The original region is not silently replaced by a favorable subset.

Table XIX reproduces the original full-menu corner and its movement under finer wealth settlement. The change from 49 to 97 nodes switches the no-adjustment ranking at this corner; the adjusted ranking remains positive. The positive relative adjustment option therefore survives a perturbation under which adjustment ceases to be necessary for positive risk. This distinction is not a failure of the finite-array inequality. It identifies the part of the economic conclusion that depends on the settlement lottery.

The common-menu and prolonged-full experiments give the same qualitative result. Their menus are not silently equated: interpolation of frozen proposals at new nodes can add controls beyond the common mesh. The common-menu arm isolates the wealth-lottery change without that additional policy-menu change. All four original corners and the center are retained for every grid and both menus, with 120 class values and selected-policy replays. At the center, the no-adjustment difference remains negative on

TABLE XX
LOCAL INDIFFERENCE BRACKETS WITH THE CALENDAR HELD FIXED

Wealth nodes	λ	Adjusted contract bracket	No-adjustment bracket
49	0	[0.370068, 0.370117]	[0.558887, 0.558936]
49	0.125	[0.339551, 0.339600]	[0.515869, 0.515918]
49	0.25	[0.309082, 0.309131]	[0.472998, 0.473047]
97	0	[0.320166, 0.320215]	[0.513477, 0.513525]
97	0.125	[0.288135, 0.288184]	[0.467773, 0.467822]
97	0.25	[0.256055, 0.256104]	[0.422266, 0.422314]
145	0	[0.302783, 0.302832]	[0.497852, 0.497900]
145	0.125	[0.270313, 0.270361]	[0.451367, 0.451416]
145	0.25	[0.237646, 0.237695]	[0.405078, 0.405127]

All brackets use the full prolonged menu. Bisection begins on $[0.2, 0.6]$ and retains every evaluated difference. Opposite signs and continuity locate a crossing; global uniqueness is not inferred. Changing the wealth lottery shifts both boundaries without changing the one-eighth-year commitment interval.

all three wealth grids. Thus the evidence maps the movement of the boundary rather than asserting that the opposite-choice phenomenon disappears everywhere.

The indifference brackets in Table XX explain that movement on the contract scale. Both regime boundaries shift as the wealth lottery is changed, and the interval between them remains economically nonempty at the inspected mixture weights. The full solver trace records all bisection evaluations. These are local brackets, not a proof of global monotonicity of the difference of two convex value functions.

TABLE XXI
A COMMON NESTED DECISION REGION ACROSS THREE SETTLEMENT PROTOCOLS

Wealth nodes	Lower Δ^{adj}	Upper Δ^0	Both signs certified
49	6.8695×10^{-5}	-2.0913×10^{-4}	Yes
97	1.7639×10^{-4}	-9.4581×10^{-5}	Yes
145	2.1143×10^{-4}	-5.7401×10^{-5}	Yes

The same closed region is $[0, 0.125] \times [0.4, 0.425]$. Bounds shown use the full-menu bank and corrected chord. Count and mesh-only lower-bank checks are also retained. This is a continuum certificate in (λ, d) for each of three specified finite protocols, not a continuum claim over every wealth grid or the limiting diffusion. The region was selected after inspecting the earlier robustness results; it was not preregistered.

Table XXI supplies an additional regional statement. It uses one common nested rectangle and separately certifies its two signs under each of the three full-menu finite protocols. The old rectangle and its failed corner are retained in Table XIX; the nested region is not presented as an unannounced replacement. Its validity over intermediate contracts and mixture probabilities comes from tensor coefficient bounds, not from testing a few parameter points. The discrete set of settlement protocols is itself not promoted to a continuum of all possible discretizations.

12.2. Which Primitive Supports the Dynamic Option?

The dynamic counterfactuals distinguish the reference-state payoff from ordinary risk aversion. At the center, the baseline relative adjustment option is approximately

TABLE XXII
DYNAMIC MECHANISMS AT THE ORIGINAL REGION CENTER

Intervention	$10^4 \Delta^{\text{adj}}$	$10^4 \Delta^0$	10^4 relative option
Baseline	1.9674	-2.0933	4.0607
Tilt $a = -0.25$	1.5564	-1.3521	2.9085
Tilt $a = 0.25$	2.3858	-2.8501	5.2359
Tilt $a = 1$	3.8022	-5.1768	8.9790
Normalization $s = 0$	9.1981	9.1981	0.0000
Normalization $s = 0.5$	11.6551	10.5243	1.1309
Normalization $s = 1.5$	-8.9524	-15.8240	6.8716
Liquidation $z = 0$	-20.0244	-24.7728	4.7484
Liquidation $z = 2$	21.1887	17.8547	3.3340
Zero-covariance law	4.1345	0.9752	3.1593

Except for the named intervention, parameters are the original $(\lambda, d, k) = (0.125, 0.425, 2)$. Every entry reoptimizes both risk classes in both adjustment regimes. The normalization multiplier changes $b_0(u) = 1/(1 - u)$, not marginal consumption utility. Values shown as zero at $s = 0$ are numerically negligible, not an asserted exact zero theorem. These model counterfactuals are not empirically identified causal effects.

4.061×10^{-4} . Removing $b_0(u)$ while preserving marginal consumption utility makes this difference numerically negligible and leaves positive risk in both regimes. Increasing its multiplier to 1.5 instead gives a larger positive relative option while making both risk rankings nonpositive. The cardinal value of preference states is therefore a principal economic primitive of the dynamic example, not an innocuous normalization.

The common tilt illustrates why the two-stage result is not a calibration of the full economy. A tilt of one reverses the relative option in the two-stage example, whereas in the dynamic economy it increases the positive relative option at the inspected center. In the latter experiment the no-adjustment value changes because preferences continue to fluctuate, and consumption, continuation, and liquidation opportunities are endogenous. The uniform primitive theorem remains valid on its stated family; the dynamic calculation supplies the separate evidence needed for this economy.

Liquidation and covariance also move the choice boundary. Eliminating the liquidation payoff makes both rankings nonpositive while leaving a positive relative adjustment option. Doubling that payoff makes both rankings positive. The zero-covariance law gives positive risk in both regimes. These interventions retain the controls, deliberate-adjustment restriction, and positive-duration decision definition. They reveal interactions in the full model rather than crediting every change in risk choice to preference formation alone.

The optimized-moment account in Table XXIII puts the baseline magnitudes in common cardinal units. The preference-state payoff contributes about 5.093×10^{-4} to the relative option, partly offset by consumption exposure, duration, effort, and liquidation terms. This is an exact accounting across four separately optimized policies, not a causal identification theorem. The deposited counterfactuals additionally evaluate the original policies under each new primitive and report their reoptimization gains. They make clear which changes are valuation of a fixed exposure and which require altered behavior.

TABLE XXIII

OPTIMIZED-MOMENT ACCOUNTING FOR THE RELATIVE ADJUSTMENT OPTION

Component	Contribution in baseline utility units
Consumption exposure	-5.1603×10^{-6}
Cardinal preference-state payoff	5.0928×10^{-4}
Tilt exposure at baseline	0
Operating duration	-1.3726×10^{-5}
Adjustment effort	-4.9862×10^{-5}
Liquidation payoff	-3.4460×10^{-5}

The sum is the baseline relative option. Each moment is evaluated at its own class-regime optimizer. This accounting does not assign invariant causal shares; policy changes and primitive interactions are reported separately through reoptimization.

TABLE XXIV

A COMMON-TARGET ECONOMIC DECISION CERTIFICATE

Upper	Lower bank	Lower Δ^{adj}	Upper Δ^0	Upper sec.	Total sec.
Chord	Full	6.8695×10^{-5}	-5.2801×10^{-5}	2.02	10.60
Count	Full	6.8695×10^{-5}	-5.2801×10^{-5}	7.89	16.47
Chord	Mesh only	6.8695×10^{-5}	-5.2801×10^{-5}	2.02	13.79

The target is the original 1,568-action economy on $[0, 0.25] \times [0.4, 0.45]$. Every arm is charged for kernel and proposal loading, full-menu upper-anchor optimization, feasible-policy evaluation, the arithmetic audit, and independent replay. The mesh-only bank additionally pays for its own lower-policy construction. Common setup is measured once and charged identically; these are serial single-pass measurements, not independent-machine medians. The per-class arithmetic allowance is 10^{-7} .

TABLE XXV

INITIAL OPTIMIZERS BESIDE THE ORIGINAL DECISION CERTIFICATE

Regime	Class	Value	c	θ	π
Adjustment	+	-0.759029726	0.8	0.2	0.8
Adjustment	-	-0.759226466	0.8	0.2	-0.5
No adjustment	+	-0.794487049	0.8	0.0	0.8
No adjustment	-	-0.794277722	0.8	0.0	-0.5

Original center $(\lambda, d) = (0.125, 0.425)$, noncancellable mandate, complete finite menu. Both the long permission 0.8 and the short permission 0.5 bind. Consumption is also at its upper limit; deliberate adjustment is at its upper limit when permitted. The nonpositive position is risky, and the no-adjustment regime retains stochastic preferences.

12.3. Attributable Computation at the Economic Accuracy Target

Table XXIV compares streaming chord and count implementations on the original 1,568-action target and the original rectangle. Both use the same full-menu upper anchors and independently evaluated class policies. Both certify the two risk rankings. The count coefficients are weakly tighter, but the chord requires less upper construction work; in this experiment its tighter competitor does not change the accepted economic conclusion. The total includes model construction, frozen-proposal loading, endpoint optimization, lower evaluation, the arithmetic audit, and independent selected-control

replay. The gain is therefore attached to the economic decision tolerance rather than an unrelated 10^{-3} welfare target.

The mesh-only lower-bank arm is evaluated against the same full-menu upper target. It also certifies both signs. Its separately charged construction is a diagnostic of attribution, not a claim to be the most efficient non-neural solver. No network is trained for these new arms: the three historical proposals define part of their common, already deposited target. Their loading and extension are charged in construction; historical network training is a sunk input reported in the retained earlier experiments, not silently relabeled as a new training speedup. Adding an identical historical construction charge to all arms changes total percentages, but not the measured difference between these upper components.

Streaming addresses the lower-bank storage issue directly. For the baseline focal-state task, sixteen policies require 827,904 bytes of control indices, and their retained intercept–duration coefficients require 2,304 bytes. The kernel and reward payload is 775,469,952 bytes. These payloads exclude common temporary arrays and process overhead; the evidence records process peak resident memory separately, without attributing a shared process peak to one upper arm. The largest memory object remains the transition representation, not the upper coefficient correction. The streamed focal certificate does not claim to preserve simultaneous access to all dates’ and states’ lower coefficients at the same storage cost.

The new dense audit checks 60 models, 240 class recursions, and 720 parameter-point comparisons, including first-action and no-adjustment restrictions. Full-model policies are also replayed through the original selected-control transition formula. All executed values, upper and lower coefficients, source hashes, and stage timings accompany the manuscript. The bounds concern the specified finite protocols and their documented arithmetic model. No empirical replay discrepancy is used as a substitute for the additional target-error control in (76).

13. THE PRICE OF COMMITMENT AND THE LOCATION OF THE RISK BOUNDARY

The following experiments change surrender eligibility or first-date investment permissions while preserving the original economic experiment as a reported comparison. All operating classes are reoptimized under the relevant changed primitive. The source programs use the hash-pinned transition constructor, complete later menus, and frozen proposal files of the preceding experiment. No network is retrained, no historical scientific input is replaced, and no settlement-grid result is silently reassigned to a different target.

13.1. *Participation Is Quantitatively Material*

At the original center $(\lambda, d) = (0.125, 0.425)$, immediate liquidation pays $G(x_0) = 0.0223143551$. The noncancellable adjusted optimum is -0.7590297258 , whereas the no-adjustment optimum is -0.7942777218 . Thus the minimum signing grants, before any capacity carrying cost, are 0.7813440809 and 0.8165920769. Adjustment reduces the compensation required to undertake the mandate by 0.0352479960. The signing grant is not a wealth injection into the portfolio problem; its separate-account assumption is essential to these numbers.

Table XXVI also relates the grant to the amount of service actually delivered. With noncancellable operation, the principal’s break-even service flow is lower under adjustment. The calculation does not impose that benefit on the principal: below the

TABLE XXVI
PARTICIPATION AND SERVICE PROCUREMENT AT THE ORIGINAL CENTER

Contract	Regime	Duration	Grant q^*	$b_{\min}$
$F = 0.00, m = 1$	Both	0.124688	0.110570	0.886775
$F = 0.70, m = 1$	Adjustment	0.684865	0.775123	1.131790
$F = 0.70, m = 1$	No adjustment	0.436339	0.789425	1.809203
$F = 0.85, m = 1$	Adjustment	0.890233	0.781344	0.877685
$F = 0.85, m = 1$	No adjustment	0.887898	0.816592	0.919691
$m = 8$	Adjustment	0.890233	0.781344	0.877685
$m = 8$	No adjustment	0.887898	0.816592	0.919691

The agent chooses the better initial class in each regime. Grants and charges use a separate additive utility numeraire, not managed wealth. Here $C(E) = 0$; for positive capacity cost add $C(E)$ to the grant and divide the added amount by duration for the change in $b_{\min}$. The principal obtains the reported flow only during operation. These are not empirical monetary compensating variations.

displayed threshold, the specified service procurement is not mutually acceptable at the minimum grant. Free surrender after the first interval requires less compensation because it delivers much less operation. It is not a cheaper way of obtaining the same service. For a positive carrying cost $C(E)$, add $C(E)$ to the grant and $C(E)/\mathcal{A}_{p^*}$ to the break-even service flow.

A concrete positive-cost procurement example illustrates the participation margin. Take available capacity at least 0.85, use $F = E = 0.85$, let $C(E) = 0.01$ at that capacity, and let the principal's net service flow be $b = 0.90$. With adjustment the minimum grant is 0.7913441 and the principal's surplus is 0.0098659. Without adjustment the minimum grant is 0.8265921 and the principal's surplus is -0.0274839 . Thus the adjusted mandate is mutually acceptable while the same service-price and capacity terms cannot support the unadjusted mandate. This is a constructive theoretical example in the stated separate-account economy, not an empirical calibration or an optimality claim about the mandate menu.

13.2. A Finite Surrender Charge and a Nonmonotone Relative Option

At zero charge with the first interval compulsory, all four optimized policies surrender at the first eligible date. The positive and nonpositive values are respectively -0.0882558999182 and -0.0887688376826 in both regimes. The relative adjustment option is zero. The positive initial action is $(0.8, 0, 0.8)$; the nonpositive action is $(0.8, 0, -1/15)$ in the original finite menu. That menu does not contain zero risky share, and the negative position is not described as risk-free. An independent date-zero-surrender calculation instead gives $G(x_0)$, as predicted by the supersolution argument.

This free-surrender result is not inferred from a few optimized rows alone. At all eight dates, the all-action endpoint check at $d = 0.45$ has largest interior residual -0.0214966453 for the operating backup of G . Affinity in λ and nonnegative duration extend the check to the original rectangle. The boundary residual is zero. The continuous benchmark's previously established supersolution inequality has the same economic implication for its own free-stopping problem; this is not a convergence argument between the two models.

At the intermediate charge $F = 0.70$, the relative option is *negative*, approximately -0.00341186 . Adjustment still improves each individual class value. What changes is

TABLE XXVII
SURRENDER CHANGES BOTH THE LEVEL AND THE SIGN OF THE RELATIVE OPTION

m	F	$10^4 \Delta^{\text{adj}}$	$10^4 \Delta^0$	$10^4 \mathcal{O}$
1	0.00	5.1294	5.1294	0.0000
1	0.40	5.1294	5.1294	0.0000
1	0.70	43.0565	77.1751	-34.1186
1	0.80	1.9674	-1.9186	3.8860
1	0.85	1.9674	-2.0933	4.0607
2	0.00	5.4927	5.4008	0.0919
4	0.00	5.9646	5.9646	0.0000
6	0.00	22.4187	23.0450	-0.6263
7	0.00	25.6088	26.0814	-0.4726
8	0.00	1.9674	-2.0933	4.0607

Original center, full finite operating menu, four reoptimized class values per row. The first interval remains compulsory. $m = 8$ is the original mandate and has no eligible surrender date. Absolute values are monotone in F and m in the required directions; the displayed differences need not be. All 19 center specifications remain in the numerical deposit.

the class in which it is most valuable. Discounted surrender exposures are 0.248753 and 0.204167 in the adjusted positive and nonpositive classes, compared with 0.560607 and 0.533895 without adjustment. The two regimes therefore value the termination margin differently. These exposures enter (81); they are not a causal decomposition obtained by holding the optimized policy fixed while claiming it remains optimal everywhere.

The charge 0.80 restores opposite rankings at the center, but not the original uniform conclusion. The positive no-adjustment policy still uses the surrender option, with discounted incidence 0.0689951. A regional certificate at that charge fails to make its upper class difference negative. At $(\lambda, d) = (0, 0.4)$, direct reoptimization gives a positive no-adjustment difference 0.00139238. The deposit retains this adverse corner as well as the failed certificate. Raising the charge to 0.85 restores the original values and actions at the center. The uniform result below establishes more than that pointwise observation.

Minimum operating terms provide another way of changing the obligation. With free surrender after two, four, six, or seven intervals, both classes continue to prefer the positive position at the center in both regimes. The relative option is not monotone in the minimum term: for example, it is approximately -0.0000473 at seven intervals, and becomes 0.0004061 under the original eight-interval mandate. Absolute class values have the required monotonicity in the fee and the minimum term. Their differences do not inherit it.

13.3. An Enforcement–Commitment Frontier on the Original Region

The reachable supports contain 30 nodes after one interval and 735 after seven intervals. Applying Proposition 22 with the two feasible no-adjustment lower policies gives the sufficient regional charges in Table XXVIII. For surrender available after the first interval, the computed upper bound is 0.8348592806; the displayed sufficient charge is rounded upward. Using every grid node instead would require a sufficient bound above 2.53. Reachability therefore matters economically: the guarantee need cover the actual mandate’s possible continuation states, not unrelated initial conditions.

TABLE XXVIII
SUFFICIENT ENFORCEMENT CAPACITY AND THE MINIMUM OPERATING TERM

m	Term (years)	Reachable-state bound	All-node bound
1	0.125	0.834860	2.530979
2	0.250	0.764293	2.231418
3	0.375	0.681582	1.912037
4	0.500	0.585751	1.570892
5	0.625	0.474158	1.206311
6	0.750	0.342991	0.877726
7	0.875	0.187395	0.492947
8	1.000	0.000000	0.000000

Uniform over $(\lambda, d) \in [0, 0.25] \times [0.4, 0.45]$ and both adjustment regimes, on the original finite menu. Bounds are rounded upward to six decimals after the 10^{-7} allowance. A capacity E at least the stated bound implements the original mandatory continuation values with charge $F = E$, provided $E \leq \bar{E}$. These are sufficient statewise bounds, not the smallest initial-state fees or an optimal contract-design solution.

TABLE XXIX
THE ORIGINAL REGION WITH A FINITE SURRENDER CHARGE

Contract	$10^4 \Delta^{\text{adj}}$ bound	$10^4 \Delta^0$ bound	Both signs
$m = 8$	[0.6869, 3.2473]	[-3.6968, -0.5280]	Yes
$F = 0.80, m = 1$	[0.6869, 3.2473]	[-3.6968, 13.9258]	No
$F \in [0.85, 0.90], m = 1$	[0.6869, 3.2473]	[-3.6968, -0.5280]	Yes

Each row uses the original rectangle $[0, 0.25] \times [0.4, 0.45]$, both adjustment regimes, and four law subcells. Chord and count are executed separately and give the same displayed bounds. Every unscaled difference includes 2×10^{-7} arithmetic padding relative to the stored arrays. The fee interval uses a joint fee–benefit–law coefficient bound. The failed $F = 0.80$ row and its adverse reoptimized corner are retained.

These are upper bounds on the statewise enforcement requirement, not estimates of the smallest fee preserving just the initial class ranking. They apply to the original finite action target and the whole rectangle $[0, 0.25] \times [0.4, 0.45]$. They do not automatically cover the enlarged continuous first-date permission menus studied below. Larger minimum terms reduce the sufficient guarantee capacity. Adjustment weakly reduces the exact statewise requirement by (85); the supplement also reports the separately solved center thresholds in both regimes.

Table XXIX certifies the original opposite-position comparison for every

$$(\lambda, d, F) \in [0, 0.25] \times [0.4, 0.45] \times [0.85, 0.90], \quad m = 1. \quad (98)$$

Both streamed upper constructions pass with the stated arithmetic padding. Thus the original conclusion is implemented by a finite, explicitly priced enforcement capacity with a surrender right, not only by formally forbidding exit. The capacity remains a substantive economic restriction: participation requires its carrying cost to be funded, and the free and intermediate-charge outcomes remain different. The same original region, rather than a replacement favorable rectangle, appears in all three rows of the table.

TABLE XXX
THE TWO REGIME-SPECIFIC PERMISSION BOUNDARIES

λ	L_{adj}^*	L_0^*	S_{adj}^*	S_0^*
0.000	0.792922	0.816033	0.506281	0.486275
0.125	0.789009	0.810784	0.509481	0.490704
0.250	0.785115	0.805683	0.512575	0.495063

Here $d = 0.425$, with no voluntary surrender. The L roots hold $S = 0.5$; the S roots hold $L = 0.8$. Only the first-date risky-share permission changes. Search is exact over the breakpoint reduction conditional on 121 finite consumption–adjustment pairs and feasible original proposals. Reported roots are floating-point computations, not interval enclosures or continuously optimized consumption and preference-adjustment policies.

13.4. Both Investment-Permission Boundaries Move

The breakpoint search reproduces the original finite-menu class values at all nine inspected (λ, d) contracts to numerical precision. The maximum over continuous first-date risky shares, conditional on each finite consumption–adjustment pair, therefore does not reveal an omitted within-box improvement in these comparisons. The original optimizers are at both position limits, as shown beside the original decision table. The interior preference-hedging formula remains an interior characterization, not a formula for these constrained optimizers.

Table XXX places both switching permissions on the same scale. At the center and with $S = 0.5$, the adjusted class changes sign at $L \simeq 0.7890095$, and the no-adjustment class at $L \simeq 0.8107837$. The baseline $L = 0.8$ lies between them. Increasing it to 0.82 crosses both, so both regimes select a positive position. Holding $L = 0.8$ instead, the corresponding short-permission switches are $S \simeq 0.5094806$ and $S \simeq 0.4907037$. Increasing S from 0.5 to 0.51 crosses the adjusted boundary, so both regimes select a nonpositive position. The relative adjustment option remains positive in these two particular permission interventions. Its sign alone therefore does not determine whether adjustment changes the selected position class.

At the baseline center, the long-permission shadow values are approximately 0.0179009 with adjustment and 0.0194114 without adjustment. The short-permission shadow values are 0.0200460 and 0.0225172. These quantities explain the sensitivity in value units per unit of risky-share permission. The exact interpolation-cell derivatives agree with central finite differences to less than 10^{-10} in the reported nine-contract audit. A one- or two-percentage-point change in permissions can be large relative to an initial class-value margin near 2×10^{-4} .

The operating-benefit frontiers in Table XXXI are recomputed with continuation policies reoptimized as d changes. They are not obtained by extrapolating a fixed policy's duration. At $\lambda = 0.125$, the original local crossing pair is approximately (0.339574, 0.515873). With the long permission 0.82, it becomes (0.139861, 0.334568); with the short permission 0.51, it becomes (0.429666, 0.597689). Thus the same central benefit 0.425 can lie between, above, or below the two boundaries. The movement is consistent with the duration and permission-price identities in (87). The numerical root brackets identify local crossings; global uniqueness in d and directed-rounding enclosures of the roots are not inferred.

The economic content of the example is consequently a priced interaction between preference adjustment, operating commitment, and constrained exposure. Adjustment

TABLE XXXI
REOPTIMIZED BENEFIT CROSSINGS UNDER CHANGED PERMISSIONS

λ	L	S	d_{adj}^*	d_0^*
0.000	0.80	0.50	0.370085	0.558915
0.000	0.82	0.50	0.176862	0.386619
0.000	0.80	0.51	0.457404	0.636790
0.125	0.80	0.50	0.339574	0.515873
0.125	0.82	0.50	0.139861	0.334568
0.125	0.80	0.51	0.429666	0.597689
0.250	0.80	0.50	0.309087	0.473014
0.250	0.82	0.50	0.102936	0.282746
0.250	0.80	0.51	0.401952	0.558753

No voluntary surrender. Later-date policies are reoptimized at every trial benefit on the unchanged full menu. All reported local crossings have positive optimizing duration differences. Local numerical brackets and feature values are deposited; they do not establish global uniqueness or a directed-rounding root enclosure. The original permissions and both adverse first-date extensions are shown at every inspected law probability.

can reduce the compensation and enforcement required to operate while changing which binding permission is preferred. The sign of that change depends on the surrender margin and on the location of the two permission-sensitive value surfaces. The original cardinal-tilt theorem, the settlement-granularity evidence, and the same-target computational comparisons remain necessary complementary statements; none substitutes for these contract margins.

14. RECURSIVE UTILITY, TEMPORAL SELVES, AND DYNAMIC GAMES

14.1. *Merton and Epstein–Zin Policy Evaluation*

The smooth portfolio benchmark uses stationary infinite-horizon CRRA utility with $\rho = 0.04$, $\gamma = 2$, $\mu = 0.08$, $r = 0.02$, and $\sigma = 0.2$. With $c = mX$ and no terminal bequest, the optimal ratios are

$$\pi^* = \frac{\mu - r}{\gamma\sigma^2} = 0.75, \quad m^* = \frac{\rho + (\gamma - 1)\{r + (\mu - r)^2 / (2\gamma\sigma^2)\}}{\gamma} = 0.04125. \quad (99)$$

These are stationary quantities, not finite-horizon policies with an arbitrary terminal condition. Under the displayed optimum, the discounted negative wealth moment in the verification argument decays at rate $m^* > 0$.

For recursive utility, put $a = 1 - 1/\psi$ and $b = 1 - \gamma$. The difference-form Epstein–Zin driver on $c > 0$, $bV > 0$ is

$$f_{EZ}(c, V) = \frac{\rho}{a} \{c^a (bV)^{1-a/b} - bV\}. \quad (100)$$

The logarithmic case requires a limit rather than substitution at $a = 0$. In the numerical regime $\gamma = 5$, $\psi = 1.5$, with the same asset and discount parameters, homothetic optimization gives

$$\pi^* = 0.3, \quad m^* = \psi\rho + (1 - \psi)\{r + (\mu - r)^2 / (2\gamma\sigma^2)\} = 0.0455. \quad (101)$$

TABLE XXXII
EXECUTED HOMOTHETIC POLICY EVALUATION AND IMPROVEMENT

Model	Learned m	Learned π	Policy max. error	Relative A error
Merton	0.04124996	0.74999998	3.94e-08	9.13e-13
Epstein–Zin	0.04550010	0.30000000	1.03e-07	6.89e-12

The coefficient $A = \exp(q)$ and bounded actor parameters are learned by separated updates. Analytical values are used only to compute the displayed errors. The sign condition and normalized HJB are tested separately. These runs use homothetic features rather than an unrestricted neural value class.

For $V = AX^b/b$, the coefficient is $A = (\rho(m^*)^{a-1})^{b/a}$. Consumption utility is increasing on the declared domain. A negative value when $b < 0$ is required by that domain and is not a sign error.

We execute fixed-policy evaluation for the unknown coefficient A and separate automatic-differentiation updates for bounded consumption and portfolio parameters. The coefficient is represented by $A = \exp(q)$; it and the controls are learned, not set to their targets. The analytical ratios are used afterward for validation. Table XXXII distinguishes these homothetic-feature solves from the ordinary neural networks used for endogenous preferences.

Supplement S.4 derives the policy equations, the positive branch of the evaluation equation, and a verification class with the required transversality. For the displayed recursive optimum, the linearized discount rate is 0.1115, the relevant wealth moment grows at rate 0.066, and their difference is $0.0455 > 0$. A separate finite-horizon architecture test imposes $bV > 0$ and an exact terminal sign by a transformed output. It is a domain test, not an additional solution of a finite-horizon Epstein–Zin problem.

The role of these calculations is to test recursive value dependence and the evaluation graph. Neither a low residual nor the positivity transform is by itself a proof of recursive-utility existence for every policy. Equally, the fact that an argmax has no analytic formula does not define a fully nonlinear PDE. Semilinearity concerns dependence on the highest-order derivatives. Control of diffusion can make the HJB fully nonlinear, whereas a difficult drift-control maximization with a fixed diffusion need not do so. The FBSDE literature also contains value representations rather than exclusively Pontryagin adjoints (Han, Jentzen, and E, 2018); these methods should not be excluded by that distinction.

14.2. *Sophisticated Temporal Selves*

Let the state obey $s_{n+1} = F(s_n, c_n, \varepsilon_{n+1})$ and let $\delta = e^{-\rho\Delta}$. A sophisticated current self evaluates a one-period deviation using $\beta\delta$ on ordinary continuation utility. The equilibrium system is

$$c_n^*(s) \in \arg \max_{c \in \mathcal{C}(s)} \{u(c)\Delta + \beta\delta \mathbb{E}[V_{n+1}(F(s, c, \varepsilon))]\}, \quad (102)$$

$$V_n(s) = u(c_n^*(s))\Delta + \delta \mathbb{E}[V_{n+1}(F(s, c_n^*(s), \varepsilon))], \quad V_N = G. \quad (103)$$

The factor β belongs in the current self's improvement criterion but not in each step of the ordinary continuation evaluation. Repeating it in (103) instead defines a different discount recursion. We use the same equations in the main text, supplement, and implementation.

TABLE XXXIII

SOPHISTICATED CONSUMPTION RATIOS AND INDEPENDENT DEVIATIONS

β	c_0/w	c_1/w	c_2/w	c_3/w	Eval. residual
0.7	0.2827991	0.3401848	0.4312698	0.5978885	1.78e-15
1.0	0.2163112	0.2651939	0.3467520	0.5099987	1.78e-15

Four consumption dates and terminal log wealth, $\Delta = 1$, $\rho = 0.04$. Continuation is evaluated without multiplying by β again. Independent bounded one-shot optimizations against these frozen continuation values give no gain exceeding 10^{-9} on the tested wealth levels.

The executed example has four consumption dates, terminal log wealth, no interest, $w_{n+1} = w_n - c_n$, $0 < c_n < w_n$, and $\Delta = 1$. With $V_n(w) = A_n \log w + B_n$ and $A_N = 1$, backward induction gives

$$z_n \equiv \frac{c_n}{w_n} = \frac{1}{1 + \beta \delta A_{n+1}}, \quad A_n = 1 + \delta A_{n+1}, \quad B_n = \log z_n + \delta \{A_{n+1} \log(1 - z_n) + B_{n+1}\}. \quad (104)$$

Strict concavity establishes the one-shot optimum. We compare it with independent bounded scalar optimization and evaluate the ordinary recursion at four wealth levels. All consumption columns are ratios, not levels. The no-present-bias case $\beta = 1$ recovers exponential discounting.

A two-consumption-date example with terminal log wealth and $\delta = 1$ makes the distinction transparent: the sophisticated first ratio is $1/(1 + 2\beta)$, whereas the erroneous geometrically repeated-beta recursion gives $1/(1 + \beta + \beta^2)$. They equal 0.416667 and 0.456621 at $\beta = 0.7$. The discrepancy is economic, not a change of notation.

14.3. Stochastic Capacity Competition

For player i , evaluation freezes the policy profile, and improvement maximizes only player i 's criterion with rivals' controls detached. A dynamic equilibrium must also survive unilateral changes in the player's entire continuation policy. We therefore distinguish a one-period Hamiltonian gap from dynamic exploitability,

$$\text{Exploit}_i(s) = \sup_{\tilde{\pi}_i} J_i(\tilde{\pi}_i, \pi_{-i}; s) - J_i(\pi_i, \pi_{-i}; s). \quad (105)$$

A renewal-capacity regression has two firms, binary capacities $K_i \in \{0, 1\}$, and demand $D \in \{1, 1.3\}$ with transition matrix $\begin{pmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{pmatrix}$. Output is $q_i = K_i/3$, and revenue is $q_i(D - q_i - q_j)$. Investment $a_i \in [0, 1]$ is the probability of capacity one next period, conditional on which the firms' capacity draws are independent. Its cost is $0.4a_i^2/2$. This includes the cost of maintaining capacity; it is not a free reset. The horizon is four dates, the discount factor is 0.95, and terminal value is $0.05K_i$.

For every date and all eight states, stage best responses are clipped affine functions of the rival's investment. Their composed contraction is checked, and iteration is repeated from three initial profiles. After the equilibrium profile is computed, a separate backward best-response program freezes the rival at every date and optimizes the deviating firm's continuation. The original output compared the final, date-zero best-response value. We now compare the best-response value with the equilibrium value inside every step of the backward recursion; the all-date gain is below the stated 10^{-10} tolerance. The saved policies and values cover the complete finite state space.

TABLE XXXIV
INVESTMENT WITH PERSISTENT CAPACITIES

K_1	K_2	$a_{1,0}(D = 1)$	$a_{1,0}(D = 1.3)$	$a_{1,10}(D = 1)$	$a_{1,11}(D = 1)$
0	0	0.700239	0.816869	0.521484	0.118750
0	1	0.665094	0.812634	0.442261	0.118750
1	0	0.141975	0.163600	0.108478	0.023750
1	1	0.133116	0.162538	0.088661	0.023750

Twelve investment dates and capacity survival $s = 0.8$. Capacity-one probability is $sK_i + (1 - sK_i)a_i$. Investment therefore depends on installed capacity and on continuation opportunities. The largest independently computed full unilateral deviation gain over all 192 date–player–state combinations is 4.441×10^{-16} .

The renewal transition makes next capacity independent of current capacity, conditional on the chosen investment. Its early policy consequently has a closed form. Writing P for the demand transition matrix, the first three dates satisfy

$$a_n(D) = \frac{\delta \{\mathbb{E}[D' | D] / 3 - 1/9\}}{\kappa + \delta/9}, \quad a_3 = \frac{0.05\delta}{\kappa}. \quad (106)$$

We retain this case as a regression test and add economically persistent capital. With survival $s = 0.8$, the new twelve-date transition is

$$\Pr(K_{i,n+1} = 1 | K_{i,n}, a_{i,n}) = sK_{i,n} + (1 - sK_{i,n})a_{i,n}. \quad (107)$$

Investment creates capacity when none is installed and offsets the risk of losing an installed unit. Current capacity now changes the marginal return to investment. All other flow-profit, cost, demand-transition, and terminal-value primitives are held fixed.

The one-period best responses remain clipped affine functions of the rival’s action. The implementation enumerates all nine pairs of binding and interior constraints, rather than imposing interiority. The maximum composed best-response slope product is 0.090398, below one. Table XXXIV shows dependence on installed capacity, the rival’s capacity, demand, and the remaining horizon. A separate full best-response recursion finds a maximum unilateral dynamic gain of 4.441×10^{-16} over all 192 date–player–state combinations. Deliberately replacing one late-date action by a nonoptimal investment produces a large detected gain at a recorded date and state, checking that the diagnostic does not inadvertently reduce to date zero.

This is an independent dynamic-game reference, not a trained high-dimensional game network. A separate automatic-differentiation test verifies that the own-action gradient is present and the rival gradient is absent. The static Cournot comparison, with Nash quantities $1/3$ and joint-profit quantities $1/4$, remains a diagnostic for the distinction between equilibrium and collusion, not a substitute for the dynamic check. Supplement S.5 sets out the equilibrium operator, resource normalization for larger markets, and a conditional small-player approximation bound. Symmetry of initialization is not used as an equilibrium certificate.

The portfolio and recursive-utility benchmarks build on [Merton \(1971\)](#), [Epstein and Zin \(1989\)](#), and [Duffie and Epstein \(1992\)](#). The normalization and admissible comparison class used here are specified in Supplement S.4.

15. CONCLUSION

Neural Bellman Operators separate proposal, evaluation, improvement, and certification so that a computational policy can be used in an economic counterfactual. The substantive question is not whether a method contains a neural network, but whether its representation and its error account support the change in primitives being studied.

For rewards with fixed transitions, the policy-feature and optimistic-support geometry is established. We retain its complete finite-model implementation, action-sign exclusions, and repaired policies, while identifying precisely the part shared with scalar optimistic linear support. For changing transition laws, an endpoint chord alone is invalid. The cross-operator correction in Theorem 10, feasible Bernstein policy values, and statewise switching provide a continuum welfare guarantee that incorporates this missing effect. Approximate neural endpoints can enter through certified residual supersolutions rather than assumed exact evaluation.

In the stopped preference economy, operating benefits and liquidation levels reduce to a contract contrast. A risk-specific duration–effort ratio then characterizes regular movements of an indifference locus, and a matched restriction on deliberate preference adjustment identifies the option-value contribution to the ranking of risky positions. A primitive consumption-exposure condition supplies a positive risk-specific option bound over a probability–cost family. In the unchanged finite stopped economy, tensor Bernstein upper and lower values certify opposite risk rankings throughout a joint contract–transition region. This connects the reusable transition representation to the economic decision itself. The numerical threshold brackets retain their separate direct-solver interpretation; uniqueness and a diffusion interpretation are not inferred from a strict regional sign certificate.

The matched count comparison determines what the transition correction buys. It is not the sharpest upper oracle: localization with the same policy bank favors count information. Its gain is compressed horizon work and storage, with a total coefficient-error bound that allows policy kinks. The common-bank frontier and the autonomous refinement experiment separately report the price of precision and the demand for new endpoint solves. Lower-policy storage and exhaustive finite-action work remain in that account.

The resource comparison establishes the importance of reusable structure on both sides of a computational comparison. A nonlinear convex critic improves on the fitted quadratic continuation in the recorded experiments, but an exact reusable parametric quadratic program is faster than learned deployment for the tested short quadratic economy. Neither that fact nor the mesh-only policy-bank ablation invalidates the transfer certificate. They determine what can be attributed to learning and what should instead be attributed to policy evaluation, structural optimization, and parameter reuse.

The new economic analysis makes the decision horizon and cross-preference cardinal primitive explicit. A first-risk choice concerns a positive-duration commitment, not a control changed at a single instant. The primitive option result survives a nontrivial common-tilt family, while full-model counterfactuals show that its dynamic magnitude and its effect on risk choice depend on preference-state payoffs, liquidation, duration, and covariance. Wealth-lottery perturbations move the original indifference boundary; their failures remain alongside an additional common-region certificate. These are stronger distinctions between economic propositions, not substitutes for the target-error control needed to transfer a sign to the diffusion benchmark.

At the original finite decision target, streaming chord and count implementations certify the same risk comparison with different work. The experiment charges common

construction and validation costs, retains a mesh-only lower-policy comparator, and reports kernel, policy, and coefficient storage separately. It therefore links compression to the economic accuracy requirement without converting a component saving into a claim about neural necessity or total process memory.

Participation and enforcement provide a further economic interpretation. A separately financed mandate is acceptable only if its signing grant covers the agent's outside option and capacity cost, and if the principal values the delivered operating service sufficiently. At the original center, deliberate adjustment reduces the minimum grant by about 0.03525 utility-numeraire units. A statewide enforcement frontier shows that adjustment weakly substitutes for guarantee capacity. On the original parameter rectangle, a finite charge between 0.85 and 0.90 implements the original opposite-position result even when surrender is available after the first interval. Free surrender eliminates the relative option at the inspected initial contracts, while an intermediate charge reverses it; these adverse outcomes remain part of the same contracting analysis.

The actual financial choices bind both the long and short permissions. An exact first-date risky-share breakpoint calculation and the reoptimized benefit frontiers show how modest permission changes move both regime boundaries past the original central contract. The permission shadow values and discounted surrender exposures identify the margins behind that movement. They explain why a preference-adjustment premium, its participation value, and its effect on the chosen position class are different economic claims. The error account likewise distinguishes evaluation of the stored arrays from a real-arithmetic constructor or another economic target.

The continuous-time formulation, finite transition targets, neural proposals, recursive preferences, temporal selves, and persistent dynamic competition remain distinct parts of the framework. Keeping their assumptions and evidence explicit permits new representations and counterfactuals without changing the economic meaning of the reported welfare guarantee.

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